

# Advanced Applied Mathematics

Complete Class Notebook

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# Advanced Applied Mathematics

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## Laplace's Integral

We consider integrals of the form

$$\int_a^b f(t)e^{x\phi(t)} dt, \quad x \rightarrow \infty.$$

### Example 1: Crude Approximation

Consider the Gaussian integral

$$\int_{-\infty}^{\infty} e^{-xt^2} dt.$$

The integrand has a sharp peak at  $t = 0$ , with

$$\text{width} \approx \frac{1}{\sqrt{x}}, \quad \text{height} = 1,$$

so the integral scales as

$$I \sim O\left(\frac{1}{\sqrt{x}}\right),$$

which indicates the *order of growth rate*.

The integral evaluates exactly to

$$I = \int_{-\infty}^{\infty} e^{-xt^2} dt = \sqrt{\frac{\pi}{x}} \sim O\left(\sqrt{\frac{1}{x}}\right).$$

### Checking the Tail Contribution

To verify that the "tail" region is negligible:

$$\int_{10}^{\infty} e^{-xt^2} dt < \int_{10}^{\infty} e^{-10xt} dt = \frac{1}{10x} e^{-100x} \ll O\left(\frac{1}{\sqrt{x}}\right).$$

Thus, for  $x \rightarrow \infty$ ,

$$\int_0^b e^{-xt^2} dt \approx \int_{-\infty}^{\infty} e^{-xt^2} dt = \sqrt{\frac{\pi}{x}}.$$

**Example 2**

Consider

$$I(x) = \int_{-\infty}^{\infty} e^{-x \cosh t} dt.$$

Recall

$$\cosh t = \frac{e^t + e^{-t}}{2}.$$

For small  $t$ , we expand

$$\cosh t = 1 + \frac{t^2}{2!} + \frac{t^4}{4!} + \cdots,$$

but we do not need the full expansion because the contribution from large  $t$  is *exponentially small*.

Thus, the integral is approximated by expanding around  $t = 0$ :

$$I(x) \approx \int_{-\infty}^{\infty} e^{-x\left(1+\frac{t^2}{2}\right)} dt.$$

**Laplace's Method (General Form)**

For

$$I(x) = \int_a^b f(t) e^{x\phi(t)} dt, \quad x \rightarrow \infty,$$

if  $\phi(t)$  attains its maximum at  $t_c \in (a, b)$ , then  $e^{x\phi(t)}$  develops an infinitely sharp peak at  $t_c$ .

Only the neighborhood of  $t = t_c$  contributes to  $I(x)$  as  $x \rightarrow \infty$ .

**Example 2 (Continued)**

We start with the general form of Laplace's integral near the maximum point:

$$I(x) = \int_{t_c-\varepsilon}^{t_c+\varepsilon} f(t) e^{x\phi(t)} dt,$$

where the tails beyond  $t_c \pm \varepsilon$  do not contribute significantly as  $x \rightarrow \infty$ .

For our example

$$I(x) = \int_{-\infty}^{\infty} e^{-x \cosh t} dt, \quad x \rightarrow \infty,$$

the leading-order approximation is

$$I(x) \approx \sqrt{\frac{2\pi}{x}} e^{-x}.$$

The "bad part" (i.e., contributions from large  $t$ ) is exponentially small in the integral. To see this quantitatively, we make the substitution:

$$\cosh t = 1 + \tau^2.$$

Using the identity  $\cosh^2 t - \sinh^2 t = 1$ , we have:

$$\sinh t dt = 2\tau d\tau \quad \Rightarrow \quad dt = \frac{2\tau}{\sinh t} d\tau = \frac{\sqrt{2}}{\sqrt{1 + \tau^2/2}} d\tau.$$

Substituting into the integral:

$$I(x) = \int_{-\infty}^{\infty} e^{-x(1+\tau^2)} \cdot \frac{\sqrt{2}}{\sqrt{1+\tau^2/2}} d\tau.$$

Expanding the prefactor as a Taylor series:

$$\frac{1}{\sqrt{1+\tau^2/2}} = 1 - \frac{\tau^2}{4} + \frac{3}{32}\tau^4 + \dots$$

we get:

$$I(x) = \sqrt{2}e^{-x} \int_{-\infty}^{\infty} e^{-x\tau^2} \left(1 - \frac{\tau^2}{4} + \frac{3\tau^4}{32} + \dots\right) d\tau.$$

Evaluating the Gaussian integrals, we obtain the full asymptotic expansion:

$$I(x) \sim \sqrt{\frac{2\pi}{x}} e^{-x} \left(1 - \frac{1}{8x} + \frac{9}{128x^2} + \dots\right).$$

Note: No matter how "wrong" the  $O(\tau^4)$  terms are, they are multiplied by an exponentially small  $e^{-x\tau^2}$ . Higher-order corrections can also be derived via integration by parts.

### Example 3: Gamma Function

The Gamma function is defined as

$$\Gamma(x+1) = \int_0^{+\infty} t^x e^{-t} dt,$$

with the special case for integers

$$\Gamma(n+1) = n!.$$

This is an example of an important function "represented" by an integral, with no simple closed-form expression.

Other examples of integral-represented functions include: - Error function:

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

- Bessel function:

$$J_0(x) = \frac{1}{\pi} \int_0^\pi \cos(x \sin t) dt$$

### Asymptotics of $\Gamma(x+1)$ via Laplace's Method

We write the Gamma function in the form suitable for Laplace's method:

$$I(x) = \int_0^{+\infty} t^x e^{-t} dt = \int_0^{+\infty} e^{x \ln t} e^{-t} dt = \int_0^{+\infty} e^{x \ln t - t} dt.$$

Find the critical point by differentiating the exponent:

$$\frac{d}{dt} (x \ln t - t) = 0 \quad \Rightarrow \quad \frac{x}{t} - 1 = 0 \quad \Rightarrow \quad t_c = x.$$

Introduce the scaling substitution  $\tau = \frac{t}{x}$  to move the critical point to  $\tau_c = 1$ :

$$t = x\tau, \quad dt = x d\tau.$$

Then:

$$I(x) = \int_0^{+\infty} e^{x \ln(x\tau) - x\tau} \cdot x d\tau = x e^{x \ln x} \int_0^{+\infty} e^{x(\ln \tau - \tau)} d\tau.$$

Near  $\tau = 1$ , expand:

$$\ln \tau = (\tau - 1) - \frac{(\tau - 1)^2}{2} + \frac{(\tau - 1)^3}{3} - \dots,$$

so

$$\ln \tau - \tau = -1 - \frac{(\tau - 1)^2}{2} + \frac{(\tau - 1)^3}{3} - \dots.$$

Substituting into the integral and keeping only the leading Gaussian contribution:

$$I(x) \sim x^{x+1} e^{-x} \int_{-\infty}^{+\infty} e^{-x \frac{(\tau-1)^2}{2}} d\tau \sim x^{x+1} e^{-x} \sqrt{\frac{2\pi}{x}}.$$

## Stirling's Formula

From the asymptotic expansion of the Gamma function, we obtain Stirling's formula:

$$\Gamma(x+1) \sim x^{x+1} e^{-x} \sqrt{\frac{2\pi}{x}}.$$

For integer  $n \in \mathbb{Z}$ , this gives:

$$n! \sim n^n e^{-n} \sqrt{2\pi n}.$$

Taking the logarithm, we have:

$$\ln n! = n \ln n - n + O(\ln n).$$

## Introduction to Asymptotic Expansion

Consider the integral

$$I(x) = \int_0^\infty \frac{e^{-xt}}{1+t} dt.$$

Using the geometric series expansion

$$\frac{1}{1+t} = 1 - t + t^2 - t^3 + t^4 - \dots,$$

we substitute into the integral:

$$I(x) \sim \int_0^\infty e^{-xt} (1 - t + t^2 - t^3 + \dots) dt.$$

To evaluate the integrals, we use integration by parts repeatedly:

$$\begin{aligned}
 \int_0^\infty e^{-xt} t^n dt &= -\frac{1}{x} \int_0^\infty t^n d(e^{-xt}) \\
 &= -\frac{1}{x} t^n e^{-xt} \Big|_0^\infty + \frac{n}{x} \int_0^\infty e^{-xt} t^{n-1} dt \\
 &= \frac{n}{x} \int_0^\infty e^{-xt} t^{n-1} dt \\
 &= \frac{n(n-1)}{x^2} \int_0^\infty e^{-xt} t^{n-2} dt \\
 &= \dots = \frac{n!}{x^n} \int_0^\infty e^{-xt} dt = \frac{n!}{x^{n+1}}.
 \end{aligned}$$

Thus, the asymptotic expansion is:

$$I(x) = \sum_{n=0}^{\infty} \frac{n!}{x^{n+1}} (-1)^n = \frac{1}{x} - \frac{1}{x^2} + \frac{2!}{x^3} - \frac{3!}{x^4} + \dots.$$

This series is an *asymptotic expansion*  $\{a_n\}$ . Note that it is *divergent*:

$$a_{n+1} = (-1)^n \frac{n!}{x^{n+1}}, \quad \left| \frac{a_{n+1}}{a_n} \right| = \frac{n+1}{x} \rightarrow \infty \text{ as } n \rightarrow \infty.$$

The source of divergence is the geometric series expansion  $\frac{1}{1+t} = 1 - t + t^2 - \dots$ , which diverges for  $|t| > 1$ . Numerical example: for  $x = 10$ , the true value is 0.09156, while the partial sum gives 0.0914 with error 0.00016.

## Remainder Term in Asymptotic Expansion

The remainder  $R_N$  represents the error made by truncating the expansion after  $N$  terms. For the integral

$$I(x) = \int_0^\infty e^{-xt} \left[ 1 - t + t^2 - t^3 + \cdots + (-t)^{N-1} \right] dt + R_N,$$

where the bracket denotes the partial sum  $S_N(x)$  up to  $N$  terms. The remainder is defined as:

$$R_N = I(x) - S_N(x) = \int_0^\infty e^{-xt} \left( \frac{1}{1+t} - \left( 1 - t + t^2 - \cdots + (-t)^{N-1} \right) \right) dt.$$

Using the geometric series identity

$$1 - t + t^2 - \cdots + (-t)^{N-1} = \frac{1 - (-t)^N}{1 + t},$$

we simplify the remainder:

$$R_N = (-1)^N \int_0^\infty e^{-xt} \frac{t^N}{1+t} dt.$$

Taking absolute value and bounding  $1/(1+t) \leq 1$ :

$$|R_N| < \int_0^\infty e^{-xt} t^N dt = \frac{N!}{x^{N+1}} \equiv a_{N+1}.$$

The error is thus bounded by the magnitude of the first neglected term. For example, with  $x = 10$ :

$$a_5 = \frac{4!}{10^5} = 0.00024, \quad \text{leading to } R_4 = 0.00016.$$

## Stopping Criterion for Maximum Accuracy

The sign of  $R_N$  alternates with  $N$ :

$$R_N = (-1)^N \int_0^\infty e^{-xt} \frac{t^N}{1+t} dt.$$

The true value lies between two consecutive partial sums:

$$\begin{aligned} S_{N-1} + R_{N-1} &< I(x) < S_N + R_N, \\ \text{or } S_N + R_N &< I(x) < S_{N-1} + R_{N-1}, \end{aligned}$$

depending on the sign of  $R_N$ . The optimal stopping point is when two consecutive partial sums bracket the true value most tightly.

## Asymptotic Series Notations and Definitions

We formalize the core notations and definitions for asymptotic analysis.

### Notation: $\ll$ (Little-o, Rigorous Definition)

$$f(x) \ll g(x) \quad \text{as } x \rightarrow x_0 \quad \Longleftrightarrow \quad \lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 0.$$

This means  $f(x)$  decays much faster than  $g(x)$  in the limit.

**Notation:  $\sim$  (Asymptotic Equivalence)**

$$f(x) \sim g(x) \quad \text{as } x \rightarrow x_0 \quad \Longleftrightarrow \quad \lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 1.$$

Equivalently,  $f(x) - g(x) \ll g(x)$  as  $x \rightarrow x_0$ .

**Notation:  $O(\cdot)$  (Big-O, Weak Notation)**

$$f(x) = O(g(x)) \quad \text{as } x \rightarrow x_0 \quad \Longleftrightarrow \quad \limsup_{x \rightarrow x_0} \left| \frac{f(x)}{g(x)} \right| < \infty.$$

Example:  $\sin x - x = O(x^3)$  as  $x \rightarrow 0$ .

**Key Comments:**

- Two functions can be asymptotic without being approximately equal in a numerical sense.
- The notation  $\sim$  does not imply the symbol  $\approx$ .
- Example:  $e^x + x \sim e^x$  as  $x \rightarrow \infty$ , since  $\lim_{x \rightarrow \infty} \frac{e^x + x}{e^x} = 1$ .
- Conversely,  $\approx$  does not imply  $\sim$ . For example,  $x^2 \approx x^3$  near  $x = 0$ , but they are not asymptotic.
- By definition, a non-zero function can never be asymptotic to 0.

**Definition of Asymptotic Expansion (Poincaré, 1886)**

A function  $f(x)$  has an asymptotic expansion written as

$$f(x) \sim a_1\phi_1(x) + a_2\phi_2(x) + \cdots + a_n\phi_n(x) + \cdots \quad (x \rightarrow x_0)$$

with respect to an asymptotic sequence  $\{\phi_n(x)\}$  if:

1. The sequence is ordered such that  $\phi_{n+1}(x) \ll \phi_n(x)$  as  $x \rightarrow x_0$ .
2. For every  $N$ , the remainder satisfies:

$$R_N(x) \equiv f(x) - \sum_{n=1}^N a_n\phi_n(x) = O(\phi_{N+1}(x)).$$

This means the remainder is much smaller than the last term retained.

**Example: Asymptotic Expansion of  $I(x)$** 

For the integral

$$I(x) = \int_0^\infty \frac{e^{-xt}}{1+t} dt,$$

we have derived the series

$$I(x) \sim \sum_{n=0}^{\infty} (-1)^n \frac{n!}{x^{n+1}}.$$



The asymptotic sequence is  $\left\{\frac{0!}{x}, \frac{1!}{x^2}, \frac{2!}{x^3}, \dots\right\}$ , which satisfies  $\phi_{n+1} \ll \phi_n$ . The remainder estimate

$$|R_N| \leq \frac{N!}{x^{N+1}} = \phi_{N+1}$$

confirms this is a valid asymptotic expansion. Thus,

$$I(x) \sim \frac{0!}{x} - \frac{1!}{x^2} + \frac{2!}{x^3} - \frac{3!}{x^4} + \dots$$

## Gamma Function and Incomplete Gamma Function

Recall the definition of the Gamma function:

$$\Gamma(x+1) = \int_0^\infty t^x e^{-t} dt.$$

Consider the integral we studied earlier:

$$I(x) = \int_0^\infty \frac{e^{-xt}}{1+t} dt.$$

Make the substitution  $\tau = (1+t)x$ , which gives  $t = \frac{\tau}{x} - 1$  and  $dt = \frac{d\tau}{x}$ . The integral transforms as follows:

$$I(x) = \int_x^\infty \frac{e^{-\tau}}{1 + \left(\frac{\tau}{x} - 1\right)} \frac{d\tau}{x} = \frac{e^x}{x} \int_x^\infty \frac{e^{-\tau} x}{\tau} \frac{d\tau}{x} = e^x \int_x^\infty \tau^0 e^{-\tau} d\tau.$$

This integral is the definition of the **\*\*upper incomplete Gamma function\*\***, denoted  $\Gamma(0, x)$ :

$$I(x) = e^x \Gamma(0, x).$$

## Properties of Asymptotic Expansions

### 1. Convergent vs. Asymptotic Series

The key difference lies in the limit process:

- **Convergent series:** For a fixed  $x$ , the partial sum  $S_N(x)$  converges to  $f(x)$  as the number of terms  $N \rightarrow \infty$ .
- **Asymptotic series:** For a fixed number of terms  $N$ , the partial sum  $S_N(x)$  converges to  $f(x)$  as the parameter  $x \rightarrow x_0$ .

### 2. Uniqueness of Coefficients $a_n$

If a function  $f(x)$  has an asymptotic expansion with respect to a given asymptotic sequence  $\{\phi_n(x)\}$ , the coefficients  $a_n$  are uniquely determined by the following recursive formulas:

$$\begin{aligned} a_1 &= \lim_{x \rightarrow x_0} \frac{f(x)}{\phi_1(x)}, \\ a_2 &= \lim_{x \rightarrow x_0} \frac{f(x) - a_1 \phi_1(x)}{\phi_2(x)}, \\ a_3 &= \lim_{x \rightarrow x_0} \frac{f(x) - a_1 \phi_1(x) - a_2 \phi_2(x)}{\phi_3(x)}. \end{aligned}$$

However, the choice of the asymptotic sequence  $\{\phi_n(x)\}$  itself is not unique.

**Example:** Consider  $f(x) = \sqrt{1+x}$  as  $x \rightarrow 0$ . We can choose a non-standard asymptotic sequence, e.g.,  $\phi_1(x) = \cos x$ ,  $\phi_2(x) = \sin x$ ,  $\phi_3(x) = \sin^3 x$ .

1. **Step 1: Verify the sequence order.** As  $x \rightarrow 0$ ,  $\cos x \sim 1$ ,  $\sin x \sim x$ ,  $\sin^3 x \sim x^3$ . Thus, the sequence satisfies  $\phi_1 \gg \phi_2 \gg \phi_3$ .

2. **Step 2: Compute coefficients.**

$$\begin{aligned} a_1 &= \lim_{x \rightarrow 0} \frac{\sqrt{1+x}}{\cos x} = 1, \\ a_2 &= \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{\sin x} = \lim_{x \rightarrow 0} \frac{\frac{1}{2}x - \frac{1}{8}x^2 + \cdots}{x - \frac{1}{6}x^3 + \cdots} = \frac{1}{2}, \\ a_3 &= \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1 - \frac{1}{2}\sin x}{\sin^3 x} = \frac{1}{6}. \end{aligned}$$

### 3. Non-Uniqueness of the Expansion

As the previous example shows, different choices of the asymptotic sequence  $\{\phi_n(x)\}$  will lead to different sets of coefficients  $\{a_n\}$  for the same function.

### 4. Not All Functions Have Asymptotic Expansions

Not every function can be expanded in a given asymptotic sequence.

- Example 1:  $f(x) = 1/x$  as  $x \rightarrow 0$  cannot be expanded in powers of  $x$  ( $\{x^\alpha\}$  with  $\alpha > 0$ ).
- Example 2:  $f(x) = e^x$  as  $x \rightarrow \infty$  cannot be expanded in negative powers of  $x$  ( $\{x^{-\alpha}\}$  with  $\alpha > 0$ ).

### 5. Convergent Taylor Series as a Special Case

A convergent Taylor series is a special case of an asymptotic expansion.

$$f(x) = \sum_{n=0}^{\infty} a_n(x-x_0)^n, \quad a_n = \frac{f^{(n)}(x_0)}{n!}.$$

It satisfies the asymptotic expansion definition because the remainder  $R_N(x) = O((x-x_0)^{N+1})$ .

### 6. Transcendentally Small Terms

These are terms that decay to zero faster than any negative power law  $1/x^\alpha$  for any  $\alpha > 0$ .

$$f(x) \ll \frac{1}{x^\alpha} \quad \text{for any } \alpha > 0.$$

Example:  $e^{-x}$  as  $x \rightarrow \infty$  is transcendentally small, since

$$\lim_{x \rightarrow \infty} \frac{e^{-x}}{1/x^\alpha} = 0.$$

This means such terms are "invisible" to asymptotic power series expansions. They cannot be represented by any finite combination of powers of  $x$ .

If  $f(x) = \sum \frac{a_n}{x^n}$ , then  $f(x) + ce^{-x}$  has the same asymptotic expansion. This is because transcendently small terms like  $e^{-x}$  are invisible to power-law asymptotic series and do not affect the coefficients.

## 7. Valid vs. Invalid Operations for Asymptotic Series

Asymptotic expansions can be added, subtracted, multiplied, divided, and integrated term by term. However, **term-by-term substitution** (e.g.,  $f(g(x))$ ) and **differentiation** are not generally valid and require careful verification.

## Method of Stationary Phase

This method deals with integrals containing rapidly oscillatory terms.

### Example 1: Basic oscillatory integral

$$I(\omega) = \int_a^b \sin(\omega t) dt = -\frac{1}{\omega} \cos(\omega t) \Big|_a^b = -\frac{1}{\omega} (\cos(\omega a) - \cos(\omega b)) = O\left(\frac{1}{\omega}\right).$$

The result is bounded by  $\pm 2/\omega$ , and the error due to cancellation is proportional to the period  $T = 2\pi/\omega$ .

### Example 2: Integral with modulation

$$I(\omega) = \int_a^b f(t) \sin(\omega t) dt = O\left(\frac{1}{\omega}\right).$$

Using integration by parts:

$$I(\omega) = -\int_a^b f(t) d\left(\frac{\cos(\omega t)}{\omega}\right) = -\frac{f(t) \cos(\omega t)}{\omega} \Big|_a^b + \frac{1}{\omega} \int_a^b \cos(\omega t) f'(t) dt.$$

The first term is  $O(1/\omega)$ , and the second term is  $O(1/\omega^2)$  after a second integration by parts.

**Riemann-Lebesgue Lemma** For  $x \rightarrow \infty$ ,

$$I(x) = \int_a^b f(t) e^{ixt} dt = O\left(\frac{1}{x}\right),$$

provided  $f(t)$  is integrable (finite) on  $[a, b]$ . The phase  $xt$  is linear in  $t$ , leading to rapid oscillation.

**Generalization to arbitrary phase** Consider the integral

$$I(x) = \int_a^b f(t) e^{ix\psi(t)} dt.$$

Using integration by parts, assuming  $\psi'(t) \neq 0$ :

$$I(x) = \int_a^b f(t) d\left(\frac{1}{ix\psi'(t)} e^{ix\psi(t)}\right) = \frac{f(t)}{ix\psi'(t)} e^{ix\psi(t)} \Big|_a^b - \frac{1}{ix} \int_a^b e^{ix\psi(t)} \frac{d}{dt} \left(\frac{f(t)}{\psi'(t)}\right) dt.$$

If  $f/\psi'$  is smooth, then  $I(x) = O(1/x)$  (generalized Riemann-Lebesgue lemma).

What if  $\psi'(t_s) = 0$  for some  $t_s \in [a, b]$ ? Near  $t_s$ , the phase is no longer linear in  $t$ , so the oscillation slows down. The point  $t_s$  is called a **stationary phase point**. The main contribution to the integral comes from a small neighborhood around  $t_s$ , where the cancellation effect is weakest.

**Example** Consider

$$I(x) = \int_0^\infty \cos \left[ x \left( \frac{t^3}{3} - t \right) \right] dt.$$

The stationary phase point satisfies

$$\frac{d}{dt} \left( \frac{t^3}{3} - t \right) = t^2 - 1 = 0 \quad \Rightarrow \quad t_s = 1.$$

Expand the phase around  $t = 1$ :

$$\psi(t) = \frac{t^3}{3} - t = -\frac{2}{3} + (t-1)^2 + O((t-1)^3).$$

Thus,

$$\begin{aligned} I(x) &\sim \int_{-\infty}^{\infty} \cos \left[ x \left( -\frac{2}{3} + (t-1)^2 \right) \right] dt = \cos \left( \frac{2x}{3} \right) \int_{-\infty}^{\infty} \cos(xu^2) du + \sin \left( \frac{2x}{3} \right) \int_{-\infty}^{\infty} \sin(xu^2) du. \\ &= \sqrt{\frac{\pi}{2x}} \left( \cos \frac{2}{3}x + \sin \frac{2}{3}x \right) = O \left( \sqrt{\frac{\pi}{2x}} \right). \end{aligned}$$

Recall the standard Fresnel integral result:

$$\int_0^\infty \sin xt^2 dt = \sqrt{\frac{\pi}{2x}}.$$

**Example: Asymptotics of the Bessel Function  $J_0(x)$  as  $x \rightarrow \infty$**

We use the integral representation of the Bessel function:

$$J_0(x) = \frac{1}{\pi} \int_0^\pi \cos(x \sin t) dt = \frac{1}{\pi} \operatorname{Re} \left( \int_0^\pi e^{ix \sin t} dt \right).$$

The phase function is  $\psi(t) = \sin t$ . Find the stationary phase points by solving  $\psi'(t) = \cos t = 0$ , which gives  $t_s = \frac{\pi}{2}$  in the interval  $[0, \pi]$ .

Expand  $\sin t$  around  $t = \frac{\pi}{2}$ :

$$\sin t = \sin \frac{\pi}{2} + \cos \frac{\pi}{2} \left( t - \frac{\pi}{2} \right) - \frac{\sin \frac{\pi}{2}}{2!} \left( t - \frac{\pi}{2} \right)^2 + \cdots = 1 - \frac{1}{2} \left( t - \frac{\pi}{2} \right)^2 + \cdots.$$

The main contribution comes from a small neighborhood around  $t = \frac{\pi}{2}$ :

$$J_0(x) \sim \frac{1}{\pi} \int_{\frac{\pi}{2}-\varepsilon}^{\frac{\pi}{2}+\varepsilon} e^{ix} \left[ 1 - \frac{1}{2} \left( t - \frac{\pi}{2} \right)^2 \right] dt = \frac{e^{ix}}{\pi} \int_{-\infty}^{\infty} e^{-\frac{ix}{2} \left( t - \frac{\pi}{2} \right)^2} dt.$$

Make the substitution  $s = \sqrt{\frac{x}{2}} \left( t - \frac{\pi}{2} \right)$ , so  $dt = \sqrt{\frac{2}{x}} ds$ :

$$J_0(x) \sim \frac{e^{ix}}{\pi} \sqrt{\frac{2}{x}} \int_{-\infty}^{\infty} e^{-is^2} ds.$$

Using the Fresnel integral result  $\int_{-\infty}^{\infty} e^{-is^2} ds = \sqrt{\pi} e^{-i\frac{\pi}{4}}$ , we get:

$$J_0(x) \sim \sqrt{\frac{2}{\pi x}} e^{i\left(x - \frac{\pi}{4}\right)}.$$

To evaluate the integral  $\int_0^{\infty} e^{-is^2} ds$ , we use the analyticity of  $e^{-iz^2}$  and Cauchy's integral theorem. We deform the integration path into the complex plane, along the ray  $z = re^{i\theta}$  with  $\theta = -\frac{\pi}{4}$ :

$$\int_0^{\infty} e^{-is^2} ds = e^{-i\frac{\pi}{4}} \int_0^{\infty} e^{-r^2} dr = e^{-i\frac{\pi}{4}} \frac{\sqrt{\pi}}{2}.$$

This confirms the result  $\int_{-\infty}^{\infty} e^{-is^2} ds = \sqrt{\pi} e^{-i\frac{\pi}{4}}$ .

## Local Analysis of ODEs (Second-Order Linear)

Consider the general homogeneous second-order linear ODE:

$$y'' + P(x)y' + Q(x)y = 0.$$

The general solution is a linear combination of two linearly independent solutions  $y_1(x)$  and  $y_2(x)$ :

$$y(x) = c_1 y_1(x) + c_2 y_2(x).$$

### Classification of Points $x_0$

1. **Ordinary Point**  $x_0$ :  $P(x)$  and  $Q(x)$  are both analytic at  $x_0$ , meaning they have convergent Taylor expansions around  $x_0$ .
2. **Regular Singular Point**  $x_0$ : The point is not ordinary, but the following conditions hold:
  - $(x - x_0)P(x)$  is analytic at  $x_0$  (i.e.,  $P(x)$  behaves no worse than  $1/(x - x_0)$  near  $x_0$ ).
  - $(x - x_0)^2 Q(x)$  is analytic at  $x_0$  (i.e.,  $Q(x)$  behaves no worse than  $1/(x - x_0)^2$  near  $x_0$ ).

At a regular singular point, the ODE admits solutions of the Frobenius form:

$$y(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^{n+\alpha},$$

where  $\alpha$  is the indicial root (a real or complex constant).

3. **Irregular Singular Point**: An irregular singular point satisfies neither the ordinary nor the regular singular conditions. There is no general standard method for solving ODEs around irregular singular points.

For ODEs around ordinary points, we expand  $y(x)$  in a Taylor series and match coefficients of different powers of  $x - x_0$ . For regular singular points, we use the Frobenius expansion.

**Example 1:** Consider the ODE:

$$x^n y'' + x y' + y = 0.$$

Divide through by  $x^n$  to write it in standard form:

$$y'' + \frac{1}{x^{n-1}} y' + \frac{1}{x^n} y = 0.$$

We classify the point  $x_0 = 0$  based on  $n$ :

- If  $n = 0$ :  $P(x) = 1/x$ ,  $Q(x) = 1$ .  $x = 0$  is an ordinary point.
- If  $n = 1$ :  $P(x) = 1$ ,  $Q(x) = 1/x$ .  $x = 0$  is a regular singular point.
- If  $n = 2$ :  $P(x) = 1/x$ ,  $Q(x) = 1/x^2$ .  $x = 0$  is a regular singular point.
- If  $n \geq 3$ :  $P(x) = 1/x^{n-1}$ ,  $Q(x) = 1/x^n$ .  $x = 0$  is an irregular singular point.

**Example 2: Airy Equation at Infinity** Consider the Airy equation:

$$y'' = xy.$$

We study its behavior around  $x \rightarrow \infty$  by transforming the variable. Let  $t = 1/x$ , so  $x = 1/t$  and  $dx/dt = -1/t^2$ .

$$y' = \frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \dot{y}(-t^2),$$

$$y'' = \frac{d}{dt}(-\dot{y}t^2) \frac{dt}{dx} = (-t^2\ddot{y} - 2t\dot{y})(-t^2) = t^4\ddot{y} + 2t^3\dot{y}.$$

Substituting into the original equation:

$$t^4\ddot{y} + 2t^3\dot{y} = \frac{1}{t}y.$$

Divide by  $t^4$ :

$$\ddot{y} + \frac{2}{t}\dot{y} - \frac{1}{t^5}y = 0.$$

At  $t_0 = 0$ , this is an irregular singular point.

Around an irregular singular point, a common strategy is to use the exponential ansatz:

$$y(x) = e^{S(x)},$$

following the method of dominant balance (Liouville-Green method).

## Dominant Balance Analysis

Consider the equation  $y'' = \frac{1}{x^3}y$  as  $x \rightarrow 0^+$ . For simpler cases like  $y'' = y$ , the solution is  $y = e^x$ . For the general case  $y'' = f(x)y$ , we assume the form  $y = e^{S(x)}$ .

Differentiating:

$$y' = S'e^S, \quad y'' = S''e^S + (S')^2e^S = (S'' + (S')^2)e^S.$$

Substitute into the ODE:

$$(S'' + (S')^2)e^S = \frac{1}{x^3}e^S.$$

Cancel  $e^S$ :

$$S'' + (S')^2 - \frac{1}{x^3} = 0.$$

Assuming  $S'' \ll (S')^2$  (dominant balance), we approximate:

$$(S')^2 \sim \frac{1}{x^3} \quad \Rightarrow \quad S' \sim \pm \frac{1}{x^{3/2}}.$$

Integrating gives the leading-order behavior:

$$S \sim \pm \frac{2}{x^{1/2}} + C(x),$$

where  $C(x)$  is a slowly varying correction term,  $C(x) \ll 1/\sqrt{x}$ .

In general, if  $S' \sim 1/x^n$  as  $x \rightarrow 0^+$ , the behavior of  $S(x)$  is:

$$S(x) \sim \begin{cases} -\frac{1}{n-1}x^{-(n-1)}, & n > 1, \\ \ln x, & n = 1, \\ x^{-(1-n)}, & n < 1. \end{cases}$$

Focusing on the blow-up solution, we substitute  $S = 2x^{-1/2} + C$  into the full equation:

$$S' = -x^{-3/2} + C', \quad S'' = \frac{3}{2}x^{-5/2} + C''.$$

Substituting into  $S'' + (S')^2 - 1/x^3 = 0$ :

$$\left(\frac{3}{2}x^{-5/2} + C''\right) + \left(-x^{-3/2} + C'\right)^2 - x^{-3} = 0.$$

Expanding and balancing the dominant terms gives  $C' \sim \frac{3}{4}\frac{1}{x}$ , so the next-order correction is  $C \sim \frac{3}{4}\ln x$ . Thus, the improved expansion is:

$$S(x) \approx 2x^{-1/2} + \frac{3}{4}\ln x + D(x).$$

For the integral  $\int x^{\varepsilon-1}dx$  as  $\varepsilon \rightarrow 0$ , we have

$$\int x^{\varepsilon-1}dx = \frac{x^{\varepsilon}}{\varepsilon}.$$

Using  $x^{\varepsilon} = e^{\varepsilon \ln x} \sim 1 + \varepsilon \ln x + \dots$ , we expand

$$\frac{x^{\varepsilon}}{\varepsilon} \sim \frac{1 + \varepsilon \ln x}{\varepsilon} = \ln x + \frac{1}{\varepsilon}.$$

Continuing the dominant balance analysis for  $S(x) = 2x^{-1/2} + \frac{3}{4}\ln x + D(x)$ , substitute into the equation  $S'' + (S')^2 = 1/x^3$ :

$$D'' + (D')^2 + \frac{3}{2}\frac{D'}{x} - \frac{2D'}{x^{3/2}} - \frac{3}{16x^2} = 0.$$

Assuming  $D'' \ll D'^2$ ,  $D'^2 \ll \frac{3}{2}\frac{D'}{x}$ , and  $\frac{2D'}{x^{3/2}} \ll \frac{3}{2}\frac{D'}{x}$ , we balance the dominant terms:

$$\frac{3}{2}\frac{D'}{x} \sim \frac{3}{16x^2} \quad \Rightarrow \quad D' \sim \frac{3}{32}\frac{1}{x}.$$

Integrating gives the next correction:

$$D(x) \sim -\frac{3}{16}x^{-1/2} + \text{const.}$$

After two iterations, the correction term vanishes as  $x \rightarrow 0^+$ . The improved expansion for  $S(x)$  is:

$$S(x) = 2x^{-1/2} + \frac{3}{4}\ln x - \frac{3}{16}x^{-1/2} + \text{const.}$$

Thus, the solution is:

$$y(x) = Ce^{2x^{-1/2} + \frac{3}{4}\ln x - \frac{3}{16}x^{-1/2}} = Ce^{\frac{2}{x^{1/2}}}x^{3/4} \left(1 - \frac{3}{16}x^{1/2} + \dots\right).$$



## Perturbation Methods

Perturbation methods are used to solve algebraic and differential equations containing a small parameter.

**Example: Regular Perturbation for an Algebraic Equation** Consider the equation

$$x^2 + \varepsilon x - 1 = 0, \quad \varepsilon \ll 1.$$

The exact solution is

$$x = \frac{-\varepsilon \pm \sqrt{\varepsilon^2 + 4}}{2} = \pm 1 - \frac{\varepsilon}{2} + \frac{\varepsilon^2}{8} + \dots$$

Using the binomial expansion  $(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3!}x^3 + \dots$ , and observing that for  $\varepsilon = 0$ ,  $x = 1$ , we assume a perturbation ansatz:

$$x = 1 + \varepsilon x_1 + \varepsilon^2 x_2 + \dots$$

Plugging the ansatz  $x = 1 + \varepsilon x_1 + \varepsilon^2 x_2 + \dots$  into the original equation  $x^2 + \varepsilon x - 1 = 0$ , we collect terms by powers of  $\varepsilon$ :

-  $O(\varepsilon^0)$ :  $1^2 - 1 = 0$  (satisfied). -  $O(\varepsilon^1)$ :  $2x_1 + 1 = 0 \Rightarrow x_1 = -\frac{1}{2}$ . -  $O(\varepsilon^2)$ :  $x_1^2 + 2x_2 + x_1 = 0 \Rightarrow x_2 = \frac{1}{8}$ .

This gives the expansion for the positive root:

$$x = 1 - \frac{1}{2}\varepsilon + \frac{1}{8}\varepsilon^2 + \dots$$

For the negative root  $x = -1$ , we use the ansatz  $x = -1 + \varepsilon x_1 + \varepsilon^2 x_2 + \dots$ .

The common goal of these methods is to convert a nonlinear equation for  $x$  into a sequence of linear equations for the coefficients  $x_1, x_2, \dots$ .

A potential drawback is the need to guess the correct asymptotic sequence beforehand.

**Iteration Method** If we do not want to assume the asymptotic sequence, we can use an iterative approach. For  $x^2 = 1 - \varepsilon x$ , we start with an initial guess  $x_0 = 1$ :

$$x_1 = \sqrt{1 - \varepsilon} = 1 - \frac{\varepsilon}{2} - \frac{\varepsilon^2}{8} + \dots$$

For the next iteration,

$$x_2 = \sqrt{1 - \varepsilon x_1} = \sqrt{1 - \varepsilon \left(1 - \frac{\varepsilon}{2}\right)} = 1 - \frac{1}{2}\varepsilon + \frac{\varepsilon^2}{8} + \dots$$

Comments:

- Good: Generates the correct asymptotic sequence automatically.
- Bad: Requires increasingly more work to get higher-order corrections. Roughly speaking, the  $n$ -th iteration is needed to get the  $n$ -th term right.

Also, the iteration may not converge.

**Example 2: Singular Perturbation** Consider the equation

$$\varepsilon x^2 + x - 1 = 0.$$

At  $\varepsilon = 0$ , the equation reduces to  $x = 1$ , a single root. As  $\varepsilon \rightarrow 0$ , the equation has two roots that behave qualitatively differently.

To see how the two roots become one, we use dominant balance: 1. Assume  $\varepsilon x^2 \sim 1 \Rightarrow x \sim \pm \frac{1}{\sqrt{\varepsilon}}$ . 2. Assume  $\varepsilon x^2 \sim -x \Rightarrow x \sim -\frac{1}{\varepsilon}$ .

The exact solution is:

$$x = \frac{-1 \pm \sqrt{1 + 4\varepsilon}}{2\varepsilon} = \frac{1}{2\varepsilon} \left[ -1 \pm \left( 1 + 2\varepsilon - 2\varepsilon^2 + \dots \right) \right].$$

Expanding both roots:

$$x = \begin{cases} 1 - \varepsilon + 2\varepsilon^2 + \dots, \\ -\frac{1}{\varepsilon} - 1 + \varepsilon - 2\varepsilon^2 + \dots. \end{cases}$$

The second root "evaporates" to infinity as  $\varepsilon \rightarrow 0$ .

Singular perturbation occurs when the limit  $\varepsilon \rightarrow 0$  differs significantly from the behavior near the limit.

**Example 3: Non-Integer Power Expansion** Consider the equation

$$(1 - \varepsilon)x^2 - 2x + 1 = 0.$$

At  $\varepsilon = 0$ , we have  $x_1 = x_2 = 1$ , a double root. A naive expansion around  $x = 1$ , assuming  $x = 1 + \varepsilon x_1 + \varepsilon^2 x_2 + \dots$ , fails because the order- $\varepsilon$  term cannot be satisfied.

The exact solution is:

$$x = \frac{1 \pm \varepsilon^{1/2}}{\varepsilon} = (1 \pm \varepsilon^{1/2})(1 + \varepsilon + \varepsilon^2 + \dots) = 1 \pm \varepsilon^{1/2} + \varepsilon \pm \varepsilon^{3/2} + \varepsilon^2 \pm \dots.$$

The correct asymptotic expansion involves non-integer powers of  $\varepsilon$ .

The correct approach for the previous double-root example is to use a general expansion form:

$$x = 1 + \varepsilon^\alpha x_1 + \varepsilon^\beta x_2 + \varepsilon^\gamma x_3 + \dots,$$

where we solve for the exponents  $\alpha, \beta, \gamma$  and coefficients  $x_1, x_2, x_3$  simultaneously. More generally,

$$x = 1 + \phi_1(\varepsilon)x_1 + \phi_2(\varepsilon)x_2 + \dots,$$

where  $\{\phi_n(\varepsilon)\}$  is the asymptotic sequence.

## Perturbation Methods for Differential Equations

**Example 1: Projectile Problem** We study the motion of an object launched from the ground under gravity, and determine the time it takes to hit the ground again.

The equation of motion, considering the variation of gravity with height, is:

$$m\ddot{x} = -\frac{GMm}{(R+x)^2} = -g\frac{R^2}{(R+x)^2}m,$$

where  $g = GM/R^2$  is the gravitational acceleration at the surface. The initial conditions are  $x(0) = 0$  and  $\dot{x}(0) = v_0$ .

If we neglect the variation of gravity ( $x \ll R$ ), the equation simplifies to  $\ddot{x} = -g$ , with solution:

$$x(t) = -\frac{1}{2}gt^2 + v_0t.$$

The trajectory is a parabola, with maximum height  $v_0^2/(2g)$  and total flight time  $2v_0/g$ .

Now, we consider  $x$  to be small but not negligible compared to  $R$ , and seek higher-order corrections.

### Non-Dimensionalization

First, we identify the characteristic scales: - Characteristic time:  $t_c = \frac{v_0}{g}$  - Characteristic length:  $x_c = \frac{v_0^2}{g}$

We introduce dimensionless variables  $y$  and  $\tau$ :

$$y = \frac{x}{x_c}, \quad \tau = \frac{t}{t_c}.$$

Transforming the derivatives:

$$\frac{dx}{dt} = \frac{d(x_c y)}{d(t_c \tau)} = \frac{x_c}{t_c} \frac{dy}{d\tau}, \quad \frac{d^2x}{dt^2} = \frac{x_c}{t_c^2} \frac{d^2y}{d\tau^2}.$$

Substituting  $x_c = v_0^2/g$  and  $t_c = v_0/g$ , we get  $x_c/t_c^2 = g$ . Thus:

$$\frac{d^2x}{dt^2} = g \frac{d^2y}{d\tau^2}.$$

The original ODE becomes:

$$g \frac{d^2y}{d\tau^2} = -g \frac{R^2}{(R + x_c y)^2} = -g \frac{1}{(1 + \frac{x_c}{R} y)^2}.$$

Let  $\varepsilon = x_c/R = v_0^2/(gR)$ , a small parameter. The dimensionless ODE is:

$$y''(\tau) = -\frac{1}{(1 + \varepsilon y)^2},$$

with initial conditions  $y(0) = 0$  and  $y'(0) = 1$ .

### Regular Perturbation Method

We assume a perturbation expansion for  $y(\tau)$ :

$$y(\tau) = y_0(\tau) + \varepsilon y_1(\tau) + \varepsilon^2 y_2(\tau) + \dots$$

Substituting into the initial conditions and collecting powers of  $\varepsilon$ :

$$y(0) = y_0(0) + \varepsilon y_1(0) + \varepsilon^2 y_2(0) + \dots = 0 \implies y_0(0) = 0, \quad y_1(0) = 0, \dots$$

$$y'(0) = y'_0(0) + \varepsilon y'_1(0) + \varepsilon^2 y'_2(0) + \dots = 1 \implies y'_0(0) = 1, \quad y'_1(0) = 0, \dots$$

The initial conditions are thus entirely reflected in the lowest-order solution.

Substituting the expansion into the ODE  $y'' = -1/(1 + \varepsilon y)^2$ , and expanding the right-hand side:

$$y_0'' + \varepsilon y_1'' + \cdots = -(1 + 2\varepsilon y + 3\varepsilon^2 y^2 + \cdots) = -(1 + 2\varepsilon(y_0 + \varepsilon y_1 + \cdots) + \cdots).$$

Collecting terms by powers of  $\varepsilon$ :

-  $O(\varepsilon^0)$ :  $y_0'' = -1$ , with  $y_0(0) = 0, y_0'(0) = 1$ . The lowest-order solution is obtained by setting  $\varepsilon = 0$ :

$$y_0(\tau) = -\frac{\tau^2}{2} + \tau.$$

In dimensional form, this corresponds to  $x(t) = -\frac{1}{2}gt^2 + v_0t$ .

-  $O(\varepsilon^1)$ :  $y_1'' = 2y_0 = 2\left(-\frac{\tau^2}{2} + \tau\right) = -\tau^2 + 2\tau$ . Integrating twice and applying  $y_1'(0) = 0, y_1(0) = 0$ :

$$y_1' = -\frac{1}{3}\tau^3 + \tau^2, \quad y_1 = -\frac{1}{12}\tau^4 + \frac{1}{3}\tau^3.$$

The solution up to first order is:

$$y(\tau) = y_0 + \varepsilon y_1 = -\frac{\tau^2}{2} + \tau + \varepsilon \left( -\frac{\tau^4}{12} + \frac{\tau^3}{3} \right) + O(\varepsilon^2).$$

To find the corrected flight time  $\tau^*$ , we solve  $y(\tau^*) = 0$ . Using an algebraic perturbation ansatz:

$$\tau^* = 2 + \varepsilon \tau_1^* + O(\varepsilon^2).$$

Substituting into the equation gives  $\tau^* = 2 + \frac{4}{3}\varepsilon + O(\varepsilon^2)$ .

### Comments on Regular Perturbation

1. The leading-order solution comes from the unperturbed problem ( $\varepsilon = 0$ ), with  $\varepsilon > 0$  contributing higher-order corrections.
2. Nonlinear problems are converted into a sequence of linear problems, with higher-order terms determined by previous terms.
3. The method is powerful and can even solve ODEs without small parameters by artificially introducing  $\varepsilon$  and setting  $\varepsilon \rightarrow 1$  at the end.

**Example 2: Perturbation for an ODE Without a Small Parameter** Consider the equation  $y'' = f(x)y$  with  $y(0) = 1, y'(0) = 1$ . We introduce a small parameter  $\varepsilon$  to make the unperturbed problem solvable:

$$y'' = \varepsilon f(x)y, \quad y(0) = 1, \quad y'(0) = 1.$$

Assume the expansion  $y = y_0 + \varepsilon y_1 + \varepsilon^2 y_2 + \cdots$ . From the initial conditions:

$$1 = y_0(0) + \varepsilon y_1(0) + \cdots \implies y_0(0) = 1, \quad y_n(0) = 0 \quad (n \geq 1),$$

$$1 = y_0'(0) + \varepsilon y_1'(0) + \cdots \implies y_0'(0) = 1, \quad y_n'(0) = 0 \quad (n \geq 1).$$

-  $O(\varepsilon^0)$ : Setting  $\varepsilon = 0$  gives  $y'' = 0$ . The solution satisfying the initial conditions is:

$$y_0(x) = x + 1.$$

-  $O(\varepsilon^n)$ : The equation becomes  $y_n'' = f(x)y_{n-1}$ . We solve for  $y_n$  iteratively using the integral form:

$$y_n(x) = \int_0^x d\tau \int_0^\tau y_{n-1}(s)f(s)ds.$$

This generates a perturbation series, and it can be shown that the series remains convergent when we set  $\varepsilon = 1$  at the end.

On the other hand, consider the expansion  $y \sim y_0 + \varepsilon y_1 + \varepsilon^2 y_2 + \varepsilon^3 y_3 + \dots$ . This asymptotic expansion approaches the unperturbed solution  $y_0 = 1 + x$  as  $\varepsilon \rightarrow 0$ .

The approximation is not just valid at a single point  $x$ , but for all  $x$ , a property known as **uniform validity**.

For example, consider the function

$$f(x, \varepsilon) = e^{-\frac{x}{\varepsilon}} + x, \quad \text{with small } \varepsilon.$$

As  $\varepsilon \rightarrow 0$ ,  $f(x, \varepsilon) \rightarrow x$ . However, the quality of the approximation depends on  $x$ : - To get a good approximation at  $x = 10^{-2}$ , we need  $\varepsilon \approx 10^{-2}$ . - To get a good approximation at  $x = 10^{-6}$ , we need  $\varepsilon \leq 10^{-6}$ .

More generally, a partial sum

$$S_N(x, \varepsilon) = \sum_{n=0}^N \varepsilon^n y_n(x)$$

is said to be **uniformly asymptotic** to  $y(x, \varepsilon)$  if, for any  $\delta > 0$ , there exists an  $\varepsilon_0 > 0$  independent of  $x$ , such that

$$\left| \frac{y(x, \varepsilon) - S_N(x, \varepsilon)}{S_N(x, \varepsilon)} \right| \leq \delta$$

for all  $x$  and  $0 < \varepsilon < \varepsilon_0$ . In other words, the error can be made small for all  $x$  simultaneously by taking  $\varepsilon$  sufficiently small. This is a typical property of regular perturbation solutions.

# Advanced Applied Mathematics

Class Notebook, Pages 23–62

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## Singular Perturbation

### Example 1: Linear ODE

Consider the equation:

$$\epsilon y'' + (1 + \epsilon)y' + y = 0$$

with boundary conditions (BC):

$$y(0) = 0, \quad y(1) = 1$$

If we use regular perturbation, we expand:

$$y = y_0 + \epsilon y_1 + \dots$$

Substitute in the original eq. For  $O(\epsilon^0)$ :

$$y_0' + y_0 = 0 \implies y_0 = ce^{-x}$$

Using BC  $y_0(0) = 0 \implies y_0 = 0$ . However, the other BC  $y_0(1) = 1$  cannot be satisfied.

Notice the original equation has an exact solution. Standard for constant coefficient ODE, try exponential ansatz  $y = e^{\lambda x}$ :

$$y' = \lambda e^{\lambda x}, \quad y'' = \lambda^2 e^{\lambda x}$$

Substitute into the ODE:

$$\epsilon\lambda^2 + (1 + \epsilon)\lambda + 1 = 0$$

$$(\epsilon\lambda + 1)(\lambda + 1) = 0 \implies \lambda = -\frac{1}{\epsilon} \quad \text{or} \quad \lambda = -1$$

Thus, the general solution is:

$$y = c_1 e^{-x} + c_2 e^{-x/\epsilon}$$

Applying the BCs:

$$y(0) = 0 \implies c_1 = -c_2$$

$$y(1) = 1 \implies c_1 e^{-1} + c_2 e^{-1/\epsilon} = 1 \implies c_1 = \frac{1}{e^{-1} - e^{-1/\epsilon}}$$

So the exact solution is:

$$y(x) = \frac{e^{-x} - e^{-x/\epsilon}}{e^{-1} - e^{-1/\epsilon}}$$

Notice that  $e^{-1/\epsilon}$  is transcendentally small as  $\epsilon \rightarrow 0$ . It does not have a Taylor expansion. This is a "Singularly Perturbed" problem.

To see what  $y(x)$  converges to with fixed  $x > 0$ :

$$y(x) \xrightarrow{\epsilon \rightarrow 0} \frac{e^{-x}}{e^{-1}} = e^{1-x} \quad (\text{Outer solution})$$

But the convergence is not uniform. For uniform convergence, we want  $e^{-x/\epsilon} < \delta$ , then we need  $x > \epsilon \ln(1/\delta)$ . In other words, we need  $x > \epsilon \ln(1/\epsilon)$  for  $y(x) \rightarrow y_{\text{out}}$ .

Within the range  $0 < x < \epsilon \ln(1/\epsilon)$ , this is a region of rapid change, known as the **boundary layer** (with a fuzzy edge).

## Matched Asymptotic Expansion

We want to "glue" the inner and outer solutions to get a full consistent solution.

**Outer Solution:** (Usually easy, Regular Perturbation)

$$O(\epsilon^0): \quad y'_0 + y_0 = 0 \implies y_{\text{out}} = ce^{-x}$$

Now we only use B.C. at  $x = 1$ :

$$y_{\text{out}}(1) = 1 \implies y_{\text{out}} = e^{1-x}$$

Similarly for  $O(\epsilon^1)$ :

$$y''_0 + y'_1 + y_1 = 0 \implies e^{1-x} + y'_1 + y_1 = 0 \implies y_1 = 0$$

Similarly,  $y_n = 0$ .

**Inner Solution:** The boundary layer is very small  $\sim \epsilon$ , naturally suggesting a "stretched version" (a rescaling) of  $x$ . Let  $X = \frac{x}{\epsilon}$ .

$$y' = \frac{dy}{dx} = \frac{dY}{dX} \frac{dX}{dx} = \frac{1}{\epsilon} \frac{dY}{dX} \equiv \frac{1}{\epsilon} Y_X$$

Similarly,  $y'' = \frac{1}{\epsilon^2} Y_{XX}$ . Substitute this into the original ODE:

$$\epsilon \frac{1}{\epsilon^2} Y_{XX} + (1 + \epsilon) \frac{1}{\epsilon} Y_X + Y = 0$$

$$\frac{1}{\epsilon} Y_{XX} + \frac{1}{\epsilon} Y_X + Y_X + Y = 0$$

$$Y_{XX} + Y_X + \epsilon(Y_X + Y) = 0$$

Now we have an ODE for the inner region, which we solve via regular perturbation:

$$O(\epsilon^0): \quad Y_{0XX} + Y_{0X} = 0 \implies Y_0 = Ce^{-X} + D$$

Use the BC at  $x = 0$ :

$$Y_0(0) = 0 \implies C = -D \implies Y_{\text{in}} = C(e^{-X} - 1)$$



**Matching:** Question:  $Y_{\text{out}}$  and  $Y_{\text{in}}$  are different, how to construct a full solution? We do matching of  $Y_{\text{in}}$  and  $Y_{\text{out}}$  by setting:

$$\lim_{X \rightarrow \infty} Y_{\text{in}} = \lim_{x \rightarrow 0} y_{\text{out}}$$

$$\lim_{X \rightarrow \infty} C(e^{-X} - 1) = -C$$

$$\lim_{x \rightarrow 0} e^{1-x} = e$$

Matching determines the constant  $C \implies -C = e \implies C = -e$ . Thus,  $Y_{\text{in}} = -e(e^{-X} - 1) = e - e^{1-x/\epsilon}$ . The overlap region gives  $y_{\text{overlap}} = e$ . The matched solution is:

$$\begin{aligned} y_{\text{match}} &= y_{\text{in}} + y_{\text{out}} - y_{\text{overlap}} \\ &= e - e^{1-x/\epsilon} + e^{1-x} - e \\ &= e^{1-x} - e^{1-x/\epsilon} \end{aligned}$$

## Location & Thickness of Boundary Layer

### Example 2

$$\epsilon y'' - y' + y = 0$$

with BCs:  $y(0) = 0, y(1) = 1$ .

**Assume B.L. is at  $x = 0$ :** Outer solution: set  $\epsilon = 0 \implies -y'_0 + y_0 = 0 \implies y_0 = Ae^x$ . Use B.C. at  $x = 1$ :  $y(1) = 1 \implies Ae^1 = 1 \implies A = e^{-1}$ .

$$y_{\text{out}} = e^{x-1}$$

Inner solution: Rescale  $X = \frac{x}{\epsilon}$ .

$$\epsilon \frac{1}{\epsilon^2} Y_{XX} - \frac{1}{\epsilon} Y_X + Y = 0 \implies Y_{XX} - Y_X + \epsilon Y = 0$$

Inner problem regular perturbation, with  $Y(0) = 0$ :

$$Y = Y_0 + \epsilon Y_1 + \dots$$

$$O(\epsilon^0): \quad Y_{0XX} - Y_{0X} = 0 \implies Y_0 = B + Ce^X$$

$Y_0(0) = 0 \implies B + C = 0 \implies Y_{\text{in}} = B(1 - e^X)$ . Try matching:

$$\lim_{X \rightarrow \infty} Y_{\text{in},0} = \lim_{x \rightarrow 0} y_{\text{out}} = e^{-1}$$

However,  $Y_{\text{in}}$  blows up exponentially as  $X \rightarrow \infty$ , making it impossible to match. (We guessed the location incorrectly).

**Assume now the B.L. is at  $x = 1$ :** Outer solution:  $y_0 = ce^x$ . Use  $y_0(0) = 0$  as B.C.  $\implies c = 0$ .

$$y_{\text{out},0} = 0$$

Inner solution: Rescale  $X = \frac{1-x}{\epsilon}$ . Now  $y' = -\frac{1}{\epsilon}Y_X$  and  $y'' = \frac{1}{\epsilon^2}Y_{XX}$ .

$$\epsilon \frac{1}{\epsilon^2} Y_{XX} + \frac{1}{\epsilon} Y_X + Y = 0 \implies Y_{XX} + Y_X + \epsilon Y = 0$$

With B.C.  $Y(0) = 1$ .

$$O(\epsilon^0): \quad Y_{0XX} + Y_{0X} = 0, \quad Y_0(0) = 1 \implies Y_0 = C + De^{-X}$$

Try matching:

$$\lim_{X \rightarrow \infty} Y_{\text{in},0} = \lim_{x \rightarrow 1} y_{\text{out}} = 0 \implies C = 0$$

Since  $Y_0(0) = 1 \implies D = 1$ . So  $Y_{\text{in},0} = e^{-X}$ .

$$y_{\text{match}} = y_{\text{in}} + y_{\text{out}} - y_{\text{overlap}} = e^{-X} + 0 - 0 = e^{-(1-x)/\epsilon}$$

**More generally for ODE**

$$\epsilon y'' + a(x)y' + b(x)y = 0$$

If  $a(x) > 0$ , B.L. is at  $x = 0$ . If  $a(x) < 0$ , B.L. is at  $x = 1$ .

## Interior Layer / Corner Layer

What if  $a(x)$  changes sign? For example, consider  $x \in [-1, 1]$ . B.L. becomes an "Interior Layer".

### Example 3

$$\epsilon y'' + xy' - y = 0, \quad y(-1) = 1, \quad y(1) = 2$$

Here  $a(x) = x$ . For  $x \in [-1, 0)$ ,  $a(x) < 0 \implies$  expect B.L. at the right end ( $x = 0$ ). For  $x \in (0, 1]$ ,  $a(x) > 0 \implies$  expect B.L. at the left end ( $x = 0$ ). Thus, we expect an "Interior Layer" (or Corner Layer) at  $x = 0$ .

*Sketch: Phase plot showing boundary layer behavior meeting at origin (Corner layer) from*

$$y(-1) = 1 \text{ and } y(1) = 2.$$

**Outer solution:** Set  $\epsilon = 0 \implies xy'_0 - y_0 = 0$ . Separation of variables:

$$\frac{dy_0}{y_0} = \frac{dx}{x} \implies \ln y_0 = \ln x + \tilde{C} \implies y_0 = cx$$

Left end:  $y(-1) = 1 \implies y_{\text{out}}^L = -x$ . Right end:  $y(1) = 2 \implies y_{\text{out}}^R = 2x$ .

**Inner solution:** Rescaling  $X = \frac{x}{\delta}$ .

$$\epsilon \frac{1}{\delta^2} Y'' + \delta X \frac{1}{\delta} Y' - Y = 0 \implies \frac{\epsilon}{\delta^2} Y'' + XY' - Y = 0$$

For the inner solution, we can't discard the first term. Dominant balance requires all 3 terms to be of the same order of magnitude.

$$\frac{\epsilon}{\delta^2} \sim 1 \implies \delta = \epsilon^{1/2}$$

This is a "thick" boundary layer (since  $\epsilon^{1/2} \gg \epsilon$ ). The ODE becomes:

$$Y'' + XY' - Y = 0$$

"BC" connect to  $y^L$  and  $y^R$ :

$$X \rightarrow +\infty : \quad Y_{\text{in}} \rightarrow 2\sqrt{\epsilon}X \implies Y_X \rightarrow 2\sqrt{\epsilon}$$

$$X \rightarrow -\infty : \quad Y_{\text{in}} \rightarrow -\sqrt{\epsilon}X \implies Y_X \rightarrow -\sqrt{\epsilon}$$

By inspection,  $Y = X$  solves the inner ODE. We use the Reduction of Order method: Set  $Y = XW(X)$  and substitute into the inner ODE:

$$XW'' + (2 + X^2)W' = 0 \implies \frac{W''}{W'} = -\frac{2 + X^2}{X} = -\frac{2}{X} - X$$

Let  $W' = V$ :

$$\frac{V'}{V} = -\frac{2}{X} - X \implies V = CX^{-2}e^{-X^2/2}$$

$$W = \int_0^X t^{-2}e^{-t^2/2} dt$$

Using integration by parts:

$$Y_{\text{in}} = C_1X + C_2 \left[ e^{-X^2/2} + X\sqrt{\frac{\pi}{2}} \operatorname{erf}\left(\frac{X}{\sqrt{2}}\right) \right]$$

Matching:

$$X \rightarrow -\infty, \quad Y'_{\text{in}} \rightarrow C_1 - C_2\sqrt{\frac{\pi}{2}} = -\sqrt{\epsilon}$$

$$X \rightarrow +\infty, \quad Y'_{\text{in}} \rightarrow C_1 + C_2\sqrt{\frac{\pi}{2}} = 2\sqrt{\epsilon}$$

Solving for the constants:

$$C_1 = \frac{1}{2}\sqrt{\epsilon}, \quad C_2 = \frac{3}{\sqrt{2\pi}}\sqrt{\epsilon}$$

## Boundary Layer in Nonlinear Systems

Consider:

$$\begin{cases} \epsilon y'' = yy' - y \\ y(0) = 1, \quad y(1) = -1 \end{cases}$$

There is no theory which tells the location of the B.L. for nonlinear systems outright. We use a Geometric Method: **Phase Plane Analysis** (Standard method in analyzing dynamical systems).

1. Define  $z = y'$ , then  $y'' = z'$ .

$$\begin{cases} y' = z \\ \epsilon z' = yz - y \implies z' = \frac{1}{\epsilon}y(z - 1) \end{cases}$$

We transform the original 2nd order problem into a 1st order 2D problem. Regard  $x$  as "time", and  $(y, z)$  as the phase plane. The vector  $(y', z')$  represents the phase velocity vector, indicating a flow. Solutions are trajectories.

More generally for an nD system:

$$\vec{x}' = \vec{f}(\vec{x}) \implies x'_i = f_i(x_1, x_2, \dots, x_n)$$

*Sketch: Phase plane with horizontal lines at  $z = 1$ ,  $y = 1$ ,  $y = -1$ . Vector field arrows show flow direction. Below  $z = 1$ , trajectories form closed loops symmetric across the  $z$ -axis.*

If  $y = 0$ ,  $y' = 0$  (no horizontal velocity). If  $z = 1$ ,  $z' = 0$  (no vertical velocity). Notice if we change  $y \rightarrow -y$ ,  $y$  does not change sign,  $y, y'$  change sign.  $\implies$  time reversal symmetry.  $\implies$  trajectories are symmetric in  $y$ .  $\implies$  trajectories below  $z = 1$  are closed loops.

Solving the original ODE  $\iff$  searching for the trajectory that satisfies B.C. In this problem, we require that initially  $y(0) = 1$ , and final  $y(1) = -1$ . Geometric method  $\implies$  many possible trajectories.

**The first type:** If  $z$  is not close to 1:  $z - 1 = O(1)$ ,  $y = O(1)$ . From (\*)  $\implies z' = O(\frac{1}{\epsilon})$ . This is the "fast" region.

If  $z$  is close to 1:  $z - 1 = O(\epsilon)$ ,  $y = O(1)$ . From (\*)  $\implies z' = O(1)$ . This is the "slow" region.

*Sketch:  $y$  vs  $x$  plot showing outer solution with mild slope and a steep boundary layer transition near  $x = 1/2$ .*

Therefore, the region away from  $z = 1$  is the "fast" region. The region close to  $z = 1$  is the "slow" region.

### Comments:

1. Inner or outer refers to  $x$ , not  $(y, z)$  space. In  $(y, z)$  plane, the fast larger region happens to be the B.L. (inner region). The region close to  $z = 1$  is the slow region (outer region). 2. Unlike real flow, in the phase space of a dynamical system, density of streamlines  $\neq$  flow velocity.

Geometric picture suggests that B.L. is at  $x = 1/2$ .

**Outer soln:**  $O(\epsilon^0)$  set  $\epsilon = 0 \implies y_0(y'_0 - 1) = 0 \implies y'_0 = 1$  or  $y_0 = 0$ . Since we are in the slow region,  $y'_0 = 1 \implies y_0 = x + C$ . (The trivial solution  $y_0 = 0$  does not satisfy the B.C.)

On the left B.C.:  $y(0) = 1 \implies y_{\text{out},L} = x + 1$ . On the right B.C.:  $y(1) = -1 \implies y_{\text{out},R} = x - 2$ .

**Inner soln:** Since B.L. is at  $x = 1/2$ , apply rescaling  $X = \frac{x-1/2}{\delta}$ .  $y' = \frac{1}{\delta}Y'$ ,  $y'' = \frac{1}{\delta^2}Y''$ . Substitute into the original ODE:

$$\epsilon \frac{1}{\delta^2} Y'' = Y \frac{1}{\delta} Y' - Y$$

$$\frac{\epsilon}{\delta} Y'' = Y Y' - \delta Y$$

Dominant balance requires  $O(\frac{\epsilon}{\delta}) \sim O(1) \implies \delta = \epsilon$ .

$$Y'' = YY' - \epsilon Y$$

Regular perturbation:  $O(\epsilon^0) : Y_0'' = Y_0 Y_0'$ . Notice that  $Y_0(0) = 0$  due to symmetry (at  $x = 1/2$ ). Also notice that  $Y_0 Y_0' = \frac{1}{2}(Y_0^2)'$ .

$$\implies Y_0'' = \frac{1}{2}(Y_0^2)' \implies Y_0' = \frac{1}{2}Y_0^2 + B \quad (B < 0)$$

Let  $B = -b^2/2$ .

$$\frac{Y_0'}{Y_0^2 - b^2} = \frac{1}{2}$$

Separation of variables:

$$Y_{\text{in},0} = -b \tanh\left(\frac{b}{2}X\right)$$

**Matching:**

$$\lim_{x \rightarrow 1/2^-} y_{\text{out},L} = \lim_{X \rightarrow -\infty} Y_{\text{in},0}$$

$$\frac{1}{2} + 1 = b \times 1 \implies b = \frac{3}{2}$$

$$\therefore y_{\text{match}} = x + 1 - \frac{3}{2} \tanh\left(\frac{3}{4} \frac{x - 1/2}{\epsilon}\right) - \frac{3}{2} \dots$$

## INTRODUCTION TO WKB METHOD

Boundary Layer (BL) Theory: good when it's a *local rapid change* (local singular structure).  
WKB Theory: good when it's a *global break down* (slowly modulated oscillations, exponential).  
(Named after Wentzel, Kramers, Brillouin).

**Example:**  $\epsilon^2 y'' + y = 0$ . Generally an oscillator:

$$m\ddot{x} = -kx \implies \ddot{x} + \frac{k}{m}x = 0 \quad (\omega = \sqrt{\frac{k}{m}})$$

Here  $\omega = \frac{1}{\epsilon} \implies y \propto \sin(\frac{x}{\epsilon})$ .

More generally:  $\epsilon^2 y'' + Q(x)y = 0 \quad (\epsilon \rightarrow 0)$ . Schrödinger Equation:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \partial_x^2 \psi + V(x)\psi$$

This looks like an irregular singular point. No Taylor / Frobenius expansion. Solution behaves as  $e^{S(x)/\delta}$  (Liouville-Green).

Given the soln of a simple oscillator  $\sin(\frac{x}{\epsilon})$ , we anticipate  $y \sim e^{S(x)/\delta}$  with  $\delta$  small: similar to BL thickness. To be determined by dominant balance.

$$S(x) = S_0(x) + \delta S_1(x) + \delta^2 S_2(x) + \dots$$

$$y' = \frac{S'}{\delta} e^{S/\delta} = \left( \frac{S'_0}{\delta} + S'_1 + \dots \right) e^{S/\delta}$$

$$\begin{aligned} y'' &= \left[ \left( \frac{S''_0}{\delta} + S''_1 + \delta S''_2 \dots \right) + \left( \frac{S'_0}{\delta} + S'_1 + \delta S'_2 \dots \right)^2 \right] e^{S/\delta} \\ &= \left[ \frac{S_0'^2}{\delta^2} + \frac{1}{\delta} (S''_0 + 2S'_0 S'_1) + (S''_1 + S_1'^2 + 2S'_0 S'_2) + \dots \right] e^{S/\delta} \end{aligned}$$

Substitute  $y''$  into ODE:

$$\epsilon^2 \left[ \frac{S_0'^2}{\delta^2} + \frac{1}{\delta} (S''_0 + 2S'_0 S'_1) + \dots \right] - Q = 0$$

Dominant balance:

$$\frac{\epsilon^2}{\delta^2} S_0'^2 \sim Q \implies \delta = \epsilon$$

$O(\epsilon^0)$ :

$$S_0'^2 = Q \implies S'_0 = \pm \sqrt{Q} \implies S_0 = \pm \int_0^x \sqrt{Q} ds$$

$O(\epsilon^1)$ :

$$S''_0 + 2S'_0 S'_1 = 0 \implies S'_1 = -\frac{S''_0}{2S'_0} = -\frac{Q'}{4Q}$$

$$\implies S_1 = -\frac{1}{4} \ln(Q) + \text{const}$$

$O(\epsilon^2)$ :



$$e^{S/\epsilon} = e^{(S_0/\epsilon + S_1)} = CQ^{-1/4}e^{S_0/\epsilon}$$

$$y \approx \frac{1}{Q^{1/4}} \left( C_1 e^{\frac{1}{\epsilon} \int \sqrt{Q} dt} + C_2 e^{-\frac{1}{\epsilon} \int \sqrt{Q} dt} \right)$$

**Example: Oscillator but with  $\omega(t)$**

$$\ddot{x} + e^{-\epsilon t} x = 0 \quad (\epsilon \rightarrow 0)$$

$\omega = e^{-\epsilon t/2}$  slowly decaying  $\omega$ . For  $t \gg \frac{1}{\epsilon}$ ,  $\omega$  decays to 0, no oscillation eventually. Apply rescaling  $T = \epsilon t \implies X(T)$ .

$$\dot{x} = \epsilon \dot{X}, \quad \ddot{x} = \epsilon^2 \ddot{X}$$

ODE:  $\epsilon^2 \ddot{X} + e^{-T} X = 0$ . WKB:  $X \propto e^{S/\epsilon}$ .  $S_0'' = -e^{-T} \implies S_0 = \pm 2ie^{-T/2}$ .  $S_1' = -\frac{S_0''}{2S_0'} \implies S_1 = -\frac{1}{4} \ln(-e^{-T}) + C = \frac{T}{4} + C$ .

$$X \sim e^{T/4} e^{S_0/\epsilon} = e^{T/4} e^{\pm \frac{2ie^{-T/2}}{\epsilon}}$$

$$X = e^{T/4} \left[ c_1 \sin \left( \frac{2e^{-T/2}}{\epsilon} \right) + c_2 \cos \left( \frac{2e^{-T/2}}{\epsilon} \right) \right]$$

$$x = e^{\epsilon t/4} \left[ c_1 \sin \left( \frac{2e^{-\epsilon t/2}}{\epsilon} \right) + c_2 \cos \left( \frac{2e^{-\epsilon t/2}}{\epsilon} \right) \right]$$

(B.C. sampled ... This suggests that  $x$  depends on  $t$  through  $\epsilon t$ .)

WKB can produce BL method (in Linear problem). Previous example:  $\epsilon y'' + (1 + \epsilon)y' + y = 0$  with  $y(0) = 0, y(1) = 1$ . We have shown using B.L. method:

$$y_{\text{out}} = e^{1-x}, \quad y_{\text{in}} \sim e(e^{-x} - 1) \implies y_{\text{match}} = e^{1-x} - e^{1-x/\epsilon}$$

Now, using WKB: L-G:  $S = S_0 + \epsilon S_1 + \epsilon^2 S_2$ .

$$y' = \frac{S'}{\epsilon} e^{S/\epsilon} = \left( \frac{S_0'}{\epsilon} + S_1' + \epsilon S_2' \right) e^{S/\epsilon}$$

ODE becomes:

$$e^{S/\epsilon} \left[ \frac{S_0'^2}{\epsilon^2} + \frac{1}{\epsilon} (S_0'' + 2S_0' S_1') + \dots \right] + (1 + \epsilon) \left( \frac{S_0'}{\epsilon} + S_1' + \dots \right) + 1 = 0$$

$O(1/\epsilon)$ :

$$S_0'^2 + S_0' = 0 \implies S_0' = 0 \quad \text{or} \quad S_0' = -1$$

If  $S_0' = 0 \implies S_0 = \text{const.}$  If  $S_0' = -1 \implies S_0 = -x + \text{const.}$

$O(\epsilon^0)$ :

$$S_0'' + 2S_0'S_1' + S_0' + S_1' + 1 = 0$$

For  $S_0' = 0$ :  $0 + 0 + 0 + S_1' + 1 = 0 \implies S_1' = -1 \implies S_1 = -x$ . For  $S_0' = -1$ :  
 $0 - 2S_1' - 1 + S_1' + 1 = 0 \implies -S_1' = 0 \implies S_1 = \text{const.}$

Therefore:

$$y = c_1 e^{-x/\epsilon} + c_2 e^{-x}$$

Applying B.C. gives:

$$y = \frac{e^{-x} - e^{-x/\epsilon}}{e^{-1} - e^{-1/\epsilon}} \approx e^{1-x} - e^{1-x/\epsilon}$$

(no matching needed).

**Summary**  $\epsilon^2 y'' = Qy$ : When  $Q > 0$ :

$$y \sim \frac{1}{Q^{1/4}} \left[ c_1 e^{-\frac{1}{\epsilon} \int \sqrt{Q}} + c_2 e^{\frac{1}{\epsilon} \int \sqrt{Q}} \right]$$

When  $Q < 0$ :

$$y \sim \frac{1}{(-Q)^{1/4}} \left[ c_1 \cos \left( \frac{1}{\epsilon} \int \sqrt{-Q} \right) + c_2 \sin \left( \frac{1}{\epsilon} \int \sqrt{-Q} \right) \right]$$

## Turning Point

$Q(x_0) = 0$  for some  $x_0$ . Solution turns from wave-like to evanescent.  $1/Q^{1/4}$  diverges  $\implies$  simple WKB breaks down. Consider the simplest case, where there's only 1 turning point  $Q(x_0) = 0$  for 1  $x_0$ , and  $Q'(x_0) \neq 0$  ( $Q$  is monotonic).

$$Q(x) \approx Q'(x_0)(x - x_0) \quad \text{Leading order}$$

*Sketch: A potential curve  $Q(x)$  crossing the  $x$ -axis at  $x_0$ , representing the turning point. The solution oscillates on one side and exponentially decays on the other. WKB breaks down near  $x_0$ , forming an "Inner layer".*

Q: How does WKB break down near  $x_0$ ? **Validity of WKB:** Need  $S_0'^2 \gg \epsilon S_0''$ .  $S_0' = \pm\sqrt{Q} \implies S_0'' = \pm\frac{Q'}{2\sqrt{Q}}$ .

$$\epsilon \frac{Q'}{2\sqrt{Q}} \ll Q \implies \epsilon Q' \ll Q^{3/2}$$

For now  $Q \approx a(x - x_0) \implies Q' \approx a = O(1)$ .

$$\epsilon \ll a^{3/2}|x - x_0|^{3/2} \implies |x - x_0| \gg O(\epsilon^{2/3})$$

$\implies$  WKB fails in the neighborhood of  $|x - x_0| < O(\epsilon^{2/3})$ . This is already suggesting an "inner scale".

**Inner layer:** When close to turning point, we can apply scaling:

$$x - x_0 = \delta X$$

$$y' = \frac{1}{\delta} Y_X, \quad y'' = \frac{1}{\delta^2} Y_{XX}$$

ODE  $\implies \frac{\epsilon^2}{\delta^2} Y_{XX} = a\delta XY$ . Dominant balance  $\implies O\left(\frac{\epsilon^2}{\delta^2}\right) = O(\delta) \implies \delta = \epsilon^{2/3}$ . (Typical in turning point problems).

Let  $Z \equiv \frac{a^{1/3}(x-x_0)}{\epsilon^{2/3}} \implies Y_{ZZ} = ZY$  (Airy Equation). General Soln:  $Y = \alpha \text{Ai}(Z) + \beta \text{Bi}(Z)$ .

**Airy function:** Like Gamma, Bessel, Airy functions have integral representations:

$$\text{Ai}(Z) = \frac{1}{\pi} \int_0^\infty \cos\left(\frac{t^3}{3} + Zt\right) dt$$

$$\text{Bi}(Z) = \frac{1}{\pi} \int_0^\infty \left[ e^{-\frac{t^3}{3} + Zt} + \sin\left(\frac{t^3}{3} + Zt\right) \right] dt$$

Asymptotic behavior of Airy function can be found using stationary phase / Laplace's Method. For here, we use L-G:  $Y = e^{S(Z)}$ .  $Y'' - ZY = 0$ . Dominant balance  $\implies$  expect  $Y \sim e^S$ .  $S = \pm \frac{2}{3} Z^{3/2}$  (first order). Verify  $S'' = O(Z^{-1/4}) \ll (S')^2 = O(Z)$ .

Higher order behavior for  $Y(Z) = \alpha \text{Ai} + \beta \text{Bi}$ : As  $Z \rightarrow +\infty$ :

$$\text{Ai}(Z) \sim \frac{1}{2\sqrt{\pi}} Z^{-1/4} e^{-\frac{2}{3}Z^{3/2}} \quad (\text{decaying})$$

$$\text{Bi}(Z) \sim \frac{1}{\sqrt{\pi}} Z^{-1/4} e^{\frac{2}{3}Z^{3/2}} \quad (\text{growing})$$

As  $Z \rightarrow -\infty$ :

$$\text{Ai}(Z) \sim \frac{1}{\sqrt{\pi}} (-Z)^{-1/4} \sin \left[ \frac{2}{3}(-Z)^{3/2} + \frac{\pi}{4} \right] \quad (\text{oscillating})$$

$$\text{Bi}(Z) \sim \frac{1}{\sqrt{\pi}} (-Z)^{-1/4} \cos \left[ \frac{2}{3}(-Z)^{3/2} + \frac{\pi}{4} \right] \quad (\text{oscillating})$$

Because in the outer region, far from turning points, we can safely use WKB:

\* On the left  $x < x_0$ ,  $Q(x) < 0$ :

$$y^L = \frac{1}{(-Q)^{1/4}} \left[ C_1 \cos \left( \frac{1}{\epsilon} \int \sqrt{-Q} \right) + C_2 \sin \left( \frac{1}{\epsilon} \int \sqrt{-Q} \right) \right]$$

\* On the right  $x > x_0$ ,  $Q(x) > 0$ :

$$y^R = \frac{1}{Q^{1/4}} \left[ C_3 e^{-\frac{1}{\epsilon} \int \sqrt{Q}} + C_4 e^{\frac{1}{\epsilon} \int \sqrt{Q}} \right]$$

Matching  $y^R$  and  $Y(Z)$ : Because we want finite solution  $\implies C_4 = 0$  (no growing).  $\implies Y = \alpha \text{Ai}(Z)$ , so  $\beta = 0$ .

$$\lim_{x \rightarrow x_0^+} y^R = \lim_{Z \rightarrow \infty} Y$$

For  $Q = a(x - x_0)$ :

$$y^R = \frac{C_3}{a^{1/4}(x - x_0)^{1/4}} e^{-\frac{1}{\epsilon} \int_{x_0}^x \sqrt{a(x-x_0)} dx} = \frac{C_3}{a^{1/4}(x - x_0)^{1/4}} e^{-\frac{2}{3} \frac{a^{1/2}}{\epsilon} (x-x_0)^{3/2}}$$

$$Y = \alpha \text{Ai}(Z) \approx \frac{\alpha}{2\sqrt{\pi}} Z^{-1/4} e^{-\frac{2}{3}Z^{3/2}}$$

Substitute  $Z = \frac{a^{1/3}(x-x_0)}{\epsilon^{2/3}}$ :

$$Y \approx \frac{\alpha \epsilon^{1/6}}{2\sqrt{\pi} a^{1/12} (x - x_0)^{1/4}} e^{-\frac{2}{3} \frac{a^{1/2}}{\epsilon} (x-x_0)^{3/2}}$$

Equating the two yields:

$$C_3 = \frac{\alpha \epsilon^{1/6}}{2\sqrt{\pi} a^{-1/6}}$$

**Matching  $y^L$  and  $Y(Z)$ :**

$$\lim_{x \rightarrow x_0^-} y^L = \lim_{Z \rightarrow -\infty} Y(Z)$$

For the outer solution on the left ( $Q < 0$ ):

$$y^L = \frac{1}{(-Q)^{1/4}} \left[ C_1 \cos \left( \frac{1}{\epsilon} \int \sqrt{-Q} dx \right) + C_2 \sin \left( \frac{1}{\epsilon} \int \sqrt{-Q} dx \right) \right]$$

For the inner solution as  $Z \rightarrow -\infty$ :

$$\begin{aligned} Y &= \alpha \operatorname{Ai}(Z) \approx \frac{\alpha}{\sqrt{\pi}} (-Z)^{-1/4} \sin \left[ \frac{2}{3} (-Z)^{3/2} + \frac{\pi}{4} \right] \\ &= \frac{\alpha}{\sqrt{\pi}} (-Z)^{-1/4} \left[ \sin \left( \frac{2}{3} (-Z)^{3/2} \right) \cos \left( \frac{\pi}{4} \right) + \cos \left( \frac{2}{3} (-Z)^{3/2} \right) \sin \left( \frac{\pi}{4} \right) \right] \\ &= \frac{\alpha}{\sqrt{\pi}} (-Z)^{-1/4} \frac{1}{\sqrt{2}} \left[ \sin \left( \frac{2}{3} (-Z)^{3/2} \right) + \cos \left( \frac{2}{3} (-Z)^{3/2} \right) \right] \end{aligned}$$

By matching the coefficients of sin and cos, we require  $C_1 = C_2$ . Substituting  $-Q = a(x_0 - x)$  and  $-Z = \frac{a^{1/3}(x_0 - x)}{\epsilon^{2/3}}$ :

$$C_1 = C_2 = \frac{\alpha \epsilon^{1/6}}{\sqrt{\pi} a^{1/6} \sqrt{2}} = \frac{1}{\sqrt{2}} C_3$$

**To Summarize: Linear ODE Soln up to 1st order** For  $\epsilon^2 y'' = Qy$  with  $Q(\infty) = 0$ :

$$y^L = \frac{2C_3}{(-Q)^{1/4}} \sin \left[ \frac{1}{\epsilon} \int \sqrt{-Q} + \frac{\pi}{4} \right]$$

$$y^R = \frac{C_3}{Q^{1/4}} e^{-\frac{1}{\epsilon} \int \sqrt{Q}}$$

Near the turning point  $x_0$ :

$$y \approx \frac{2\sqrt{\pi} a^{1/6}}{\epsilon^{1/6}} C_3 \operatorname{Ai} \left[ \frac{a^{1/3}(x - x_0)}{\epsilon^{2/3}} \right]$$

## EIGENVALUE PROBLEM

A problem that has non-trivial solns only when  $E$  of the problem has special values.

**Example 1:**

$$y'' + Ey = 0$$

$$y(0) = y(\pi) = 0$$

$E$  is a parameter. Exact soln:

$$y = \sin(\sqrt{E}x)$$

To satisfy BC 1:  $y(0) = 0$  is already met. To satisfy BC 2:  $y(\pi) = 0 \implies \sin(\sqrt{E}\pi) = 0 \implies \sqrt{E} = n$  (Integer).

$$E = n^2 \quad (n = 1, 2, 3, \dots)$$

$E$  only takes special values  $1, 4, 9, \dots \implies$  **Eigenvalues**. Corresponding solns  $y_n = \sin(nx) \implies$  **Eigenfunction**.

**Example 2:** Eigenvalue can be transcendental (不能用初等函数表示, 超越的).

$$y'' + (E - x)y = 0$$

$$y(0) = y(\infty) = 0$$

Rewrite as  $y'' = (x - E)y$ . Take  $X = x - E \implies Y'' = XY$  (Airy equation).

$$y = \alpha \text{Ai}(x - E)$$

To satisfy  $y(0) = 0 \implies \text{Ai}(-E) = 0$ .

*Sketch: Graph of the Airy function  $\text{Ai}(x)$  showing oscillatory behavior for  $x < 0$  and exponential decay for  $x > 0$ . The roots on the negative  $x$ -axis ( $R_1 \approx -2.3$ ,  $R_2 \approx -4$ ,  $R_3 \approx -5.5$ ) correspond to the eigenvalues  $E$ .*

**Example 3:**

$$y'' + EQ(x)y = 0, \quad Q(x) > 0$$

$$y(0) = y(\pi) = 0$$

Q: Is this an Eigenvalue prob?  $EQ(x)$  is essentially the curvature. For large  $E$ ,  $y'' \gg 0$  or  $y'' \ll 0$ , which means it curves sharply. Solns  $y(x)$  only exist for certain  $E$ 's that make the B.C. work  $\implies$  Eigenvalue Problem.

We are going to use WKB to find approx. eigenvalues when  $E$  is large. If  $E$  is large, let  $E \equiv \frac{1}{\epsilon^2} \implies \epsilon^2 y'' + Qy = 0$  (Schrödinger eq.) Since  $Q > 0$ , WKB gives:

$$y \sim \frac{1}{Q^{1/4}} \left[ C_1 \cos \left( \frac{1}{\epsilon} \int \sqrt{Q} dx \right) + C_2 \sin \left( \frac{1}{\epsilon} \int \sqrt{Q} dx \right) \right]$$

$$\text{B.C. } y(0) = 0 \implies C_1 = 0. \text{ B.C. } y(\pi) = 0 \implies \sin \left( \frac{1}{\epsilon} \int_0^\pi \sqrt{Q} dx \right) = 0 \implies \frac{1}{\epsilon} \int_0^\pi \sqrt{Q} dx = n\pi.$$

$$E = \frac{1}{\epsilon^2} = \frac{n^2 \pi^2}{\left( \int_0^\pi \sqrt{Q} dx \right)^2}, \quad n = 1, 2, 3, \dots$$

Notice that if  $Q = 1$ , this reduces to Example 1.

## Sturm-Liouville Theory

More generally, the eigenvalue problem:

$$a(x)y'' + b(x)y' + c(x)y + d(x)Ey = 0$$

$$y(0) = y(\pi) = 0$$

can always be transformed into Sturm-Liouville form:

$$\frac{d}{dx} \left( p(x) \frac{dy}{dx} \right) + q(x)y = -Ew(x)y$$

where  $p(x) > 0$ ,  $w(x) > 0$ ,  $q(x) \leq 0$ . This guarantees many eigenvalues  $E = E_0, E_1, E_2, \dots$  that are real and positive. Eigenfunctions  $y_n = y_0, y_1, y_2, \dots$  are orthonormal with respect to the weight  $w(x)$ :

$$\int_\alpha^\beta y_n(x)y_m(x)w(x)dx = \delta_{nm}$$

(1 if  $n = m$ , 0 if  $n \neq m$ ). If we have a complete set of  $y_0, y_1, \dots$ , then we can expand any function:

$$y(x) = c_0 y_0 + c_1 y_1 + c_2 y_2 + \dots$$

where  $c_n = \int y y_n w dx$ .



## Schrödinger Equation and Turning Points

$$\epsilon^2 y'' = (V(x) - E)y, \quad y(\pm\infty) \rightarrow 0$$

*Sketch: A potential well  $V(x)$  forming a U-shape. A horizontal line represents the energy level  $E$ . The points where  $V(x) = E$  are marked as turning points A and B. The wavefunction oscillates inside the well (between A and B) and decays exponentially outside.*

$Q(x) = V(x) - E$ . Switches sign twice at A & B.

*Sketch: Magnified view of the potential well near the bottom. Points A and B are the boundaries of the classically allowed region (oscillation). Outside A and B, the wavefunction forms exponentially decaying tails.*

Solutions  $y(x)$  only exist for certain  $E$ 's when the oscillations match consistently.

**Strategy:** use WKB, solve around A and B separately, then match consistently in the region  $A < x < B$ . Let  $y^{AR}$  be the solution extending to the right from A, and  $y^{BL}$  be the solution extending to the left from B. We require  $y^{AR} \equiv y^{BL}$  in the middle.

For  $x > B$ :

$$y^{BR} \sim \frac{C_B}{(V - E)^{1/4}} e^{-\frac{1}{\epsilon} \int_B^x \sqrt{V - E} dt}$$

Using the Airy function as a connection formula across B: For  $x < B$ :

$$y^{BL} \sim \frac{2C_B}{(E - V)^{1/4}} \sin \left[ \frac{1}{\epsilon} \int_x^B \sqrt{E - V} dt + \frac{\pi}{4} \right]$$

For  $x < A$ :

$$y^{AL} \sim \frac{C_A}{(V - E)^{1/4}} e^{-\frac{1}{\epsilon} \int_x^A \sqrt{V - E} dt}$$

Using the Airy function as a connection formula across A: For  $x > A$ :

$$y^{AR} \sim \frac{2C_A}{(E - V)^{1/4}} \sin \left[ \frac{1}{\epsilon} \int_A^x \sqrt{E - V} dt + \frac{\pi}{4} \right]$$

**Match AR & BL:** The  $x$ -dependence is different in the two integrals. We must make the  $x$ -dependence agree. Rewrite the argument of the sine function in  $y^{BL}$ :

$$\begin{aligned}
& \sin \left[ \frac{1}{\epsilon} \int_x^B \sqrt{E - V} dt + \frac{\pi}{4} \right] \\
&= \sin \left[ \frac{1}{\epsilon} \int_A^B \sqrt{E - V} dt - \frac{1}{\epsilon} \int_A^x \sqrt{E - V} dt + \frac{\pi}{4} \right] \\
&= \sin \left[ \left( \frac{1}{\epsilon} \int_A^B \sqrt{E - V} dt + \frac{\pi}{2} \right) - \left( \frac{1}{\epsilon} \int_A^x \sqrt{E - V} dt + \frac{\pi}{4} \right) \right]
\end{aligned}$$

For  $y^{BL}$  to match  $y^{AR}$  exactly (up to a sign depending on  $n$ ), the sum of the phase integrals must satisfy the **matching condition**:

$$\frac{1}{\epsilon} \int_A^B \sqrt{E - V} dt + \frac{\pi}{2} = (n + 1)\pi$$

This requires  $C_B = (-1)^n C_A$ .

**Eigen condition:**

$$\int_A^B \sqrt{E - V(x)} dx = \left( n + \frac{1}{2} \right) \pi \epsilon, \quad n = 0, 1, 2, \dots$$

## Dynamical Systems and Bifurcation

**Notation:** We have introduced dynamical systems and the notation of flow in the phase plane:

$$y' = f(y, z)$$

$$z' = g(y, z)$$

Today, we will introduce the qualitative change of the "flow" in the phase space if parameters of the system are changed  $\implies$  **bifurcation**.

### 1D Systems

#### 1. Saddle-node bifurcation (鞍节分叉)

$$\dot{x} = x^2 + r$$

where  $r$  is a parameter (quadratic form). Consider fixed points:  $0 = \dot{x} = x^2 + r$ .

*Sketch: Three parabolas plotted as  $\dot{x}$  versus  $x$ . For  $r < 0$ , the parabola intersects the  $x$ -axis twice (one stable, one unstable fixed point). For  $r = 0$ , it touches at the origin (half-stable). For  $r > 0$ , it does not intersect (no fixed points).*

\*  $r < 0$ : 2 fixed points \*  $r = 0$ : 1 half-stable fixed point \*  $r > 0$ : no fixed point

When  $r$  increases to pass 0, 2 fixed points annihilate (湮灭). The name "saddle-node" comes from higher dimensions.

### Bifurcation diagram:

*Sketch: Bifurcation diagram plotting fixed points  $x^*$  against parameter  $r$ . A sideways parabola opening to the left. The upper branch is solid (stable), and the lower branch is dashed (unstable). They meet at the critical point  $r = 0, x^* = 0$ .*

Similarly, for  $\dot{x} = r - x^2$ :

*Sketch: Phase portraits (parabolas opening downwards) and Bifurcation diagram for  $\dot{x} = r - x^2$ . The bifurcation diagram is a sideways parabola opening to the right, with the upper branch stable and lower branch unstable.*

More generally,  $\dot{x} = f(x; r)$ . Near a critical point  $(x^*, r_c)$ :

$$f(x; r) \approx a(r - r_c) + b(x - x^*)^2$$

## 2. Transcritical bifurcation

$$\dot{x} = rx - x^2$$

There is always a fixed point at  $x^* = 0$ .

*Sketch: Three parabolas  $\dot{x} = rx - x^2$  for  $r < 0$ ,  $r = 0$ , and  $r > 0$ . The roots are at  $x = 0$  and  $x = r$ . They exchange stability as they pass through each other at  $r = 0$ .*

\*  $r < 0$ : 2 fixed points \*  $r = 0$ : 1 fixed point \*  $r > 0$ : 2 fixed points

When  $r$  increases to pass 0, the 2 fixed points switch stability. **Bifurcation diagram:**

*Sketch: Transcritical bifurcation diagram plotting  $x^*$  vs  $r$ . Two intersecting straight lines ( $x^* = 0$  and  $x^* = r$ ). The line  $x^* = 0$  is stable for  $r < 0$  and unstable for  $r > 0$ . The line  $x^* = r$  is unstable for  $r < 0$  and stable for  $r > 0$ .*

**Example:**

$$\dot{x} = r \ln x + x - 1$$

Notice  $x^* = 1$  is a fixed point. Let  $x = u + 1$ . Near  $x^*$ ,  $u \ll 1$ .

$$\dot{u} = r \ln(u + 1) + u \approx r \left( u - \frac{u^2}{2} \right) + u = (r + 1)u - \frac{r}{2}u^2$$

Let  $u = aw$ :

$$\dot{w} = (r + 1)w - \frac{r}{2}aw^2$$

Let  $a = \frac{2}{r}$  to obtain the normal form.

### 3. Pitchfork Bifurcation

$$\dot{x} = rx - x^3 \quad (\text{normal form})$$

Notice the  $x \rightarrow -x$  symmetry. We anticipate symmetric pairs of fixed points.

*Sketch: Three phase line diagrams for  $\dot{x} = rx - x^3$ . For  $r < 0$ , arrows point towards the single stable fixed point at  $x = 0$ . For  $r = 0$ , arrows still point to 0 but motion is slower ("critical slowing down"). For  $r > 0$ , the origin is unstable and arrows point outward towards two new stable fixed points at  $x = \pm\sqrt{r}$ .*

#### Bifurcation diagram (Supercritical Pitchfork):

*Sketch: Pitchfork bifurcation diagram. A stable branch along  $x^* = 0$  for  $r < 0$  becomes unstable for  $r > 0$ , while two new stable branches emerge for  $r > 0$  forming a pitchfork shape.*

Similarly,  $\dot{x} = rx + x^3$  gives a subcritical pitchfork.

### Imperfect Pitchfork

$$\dot{x} = s + rx - x^3$$

Now we have 2 parameters:  $r$  and  $s$ , where  $s$  represents an imperfection. To find fixed points:  $s + rx - x^3 = 0 \implies rx - x^3 = -s$ .

*Sketch: Plots of  $y = rx - x^3$  intersected by horizontal lines  $y = -s$ . For  $r > 0$ , there are local extrema. Depending on the value of  $-s$  (e.g.,  $-s > -s_c$ ,  $-s = -s_c$ ,  $-s < -s_c$ ), there may be 1, 2, or 3 fixed points.*

#### Parameter Space $(r, s)$ :

*Sketch: The  $(r, s)$  parameter space. A cusp-shaped region exists for  $r > 0$ . Inside the cusp, there are 3 fixed points. Outside, there is 1 fixed point.*

#### Bifurcation diagrams for different $s$ :

*Sketch: Three bifurcation diagrams of  $x^*$  vs  $r$ . Top ( $S = 0$ ): Perfect pitchfork. Middle ( $S > 0$ ): The upper branch is continuous and stable, while the lower branch consists of a saddle-node bifurcation. Bottom ( $S < 0$ ): The lower branch is continuous and stable, while the upper branch has a saddle-node bifurcation. For  $S \neq 0$ , there is no bifurcation at  $r = 0$ ; instead, saddle-node (SN) bifurcations appear.*

## 2D System

$$\dot{x}_1 = f_1(x_1, x_2)$$

$$\dot{x}_2 = f_2(x_1, x_2)$$

Same as  $\dot{\vec{x}} = \vec{f}(\vec{x})$ .

In a 1D system, we are interested in: (1) fixed points (2) stability

But in a 2D system, we are also interested in: (3) closed orbits (轨迹) (4) arrangement near fixed points

*Sketch: A 2D phase plane showing various trajectories. Some curve around fixed points, others form closed loops.*

## Invariant objects in 2D dynamics

**1. Fixed points** (the same as 1D but richer) Fixed points are defined by the arrangement of the trajectories:

- Node (stable/unstable)
- Saddle
- Center (neutrally stable)
- Focus / Spiral (stable/unstable)

*Sketch: Various 2D fixed point phase portraits. A stable node (arrows pointing directly inward), an unstable node (arrows pointing outward), a saddle (arrows pointing inward along one axis and outward along the other), a center (closed concentric loops), and spirals (arrows spiraling inward or outward).*

**2. Limit cycles / periodic orbits** Isolated closed trajectories in the phase plane.

*Sketch: Limit cycles. A stable limit cycle (trajectories from inside and outside spiral towards it), an unstable limit cycle (trajectories spiral away from it), and a half-stable limit cycle.*

### Example: van der Pol equation

$$\ddot{x} + (x^2 - 1)\dot{x} + x = 0$$

Here we have nonlinear damping  $(x^2 - 1)\dot{x}$  and a simple spring restoring force  $x$ . Near the origin ( $|x| < 1$ ), the damping is negative  $\implies$  small oscillations grow (excitation). Far from the origin ( $|x| > 1$ ), the damping is positive  $\implies$  large oscillations decay (dissipation). Therefore, we expect a stable limit cycle.

More generally, **Liénard's equation**:

$$\ddot{x} + f(x)\dot{x} + g(x) = 0$$

(Nonlinear damping + restoring force). **Liénard's Theorem:** If  $g$  is odd,  $f$  is even, and  $F(x) = \int_0^x f(u)du$  satisfies certain conditions, there is a unique, stable limit cycle surrounding the origin in the phase plane.

### Bifurcation in 2D

#### Saddle-node bifurcation:

$$\dot{x} = r - x^2, \quad \dot{y} = -y$$

*Sketch: Phase portraits for  $r < 0$  (no fixed points, flow moves left),  $r = 0$  (one half-stable fixed point), and  $r > 0$  (two fixed points: a saddle and a stable node).*

#### Transcritical bifurcation:

$$\dot{x} = rx - x^2, \quad \dot{y} = -y$$

*Sketch: Phase portraits for  $r < 0$ ,  $r = 0$ , and  $r > 0$ , showing a saddle and a stable node passing through each other at the origin and exchanging stability.*

#### Pitchfork bifurcation:

$$\dot{x} = rx - x^3, \quad \dot{y} = -y$$

*Sketch: Phase portraits for  $r < 0$  (one stable node at origin),  $r = 0$ , and  $r > 0$  (origin becomes a saddle, and two stable nodes emerge symmetrically).*

## Dynamics of fixed points

Consider slowly evolving fixed points.

### 1D example:

$$\frac{dy}{dt} = y - \frac{y^3}{3} - s$$

This contains a pitchfork bifurcation structure.

*Sketch: Plot of  $dy/dt$  vs  $y$  for different values of  $s$  ( $s = 0$ ,  $s = 2/3$ ,  $s = -2/3$ ). The curve shifts vertically as  $s$  changes.*

### Bifurcation diagram (Hysteresis / Catastrophe):

*Sketch: A folded S-shaped curve plotting fixed points  $y^*$  against parameter  $s$ . It shows hysteresis jumps at  $s = 2/3$  and  $s = -2/3$ . Following the stable upper branch as  $s$  increases leads to a sudden drop (catastrophe) at  $s = 2/3$ .*

In practice, the change of  $s$  is never "quasistatic". **Q:** What if  $s$  is varied at a small but finite rate? This leads to a **delayed bifurcation**.

*Sketch: Delayed bifurcation diagram. The actual state  $y(t)$  follows the stable branch but overshoots past the critical point  $s_0 = 2/3$  before jumping down at some  $s_1$ . The delay is  $s_1 - s_0$ .*

### Well-posed problem:

$$\frac{dy}{dt} = y - \frac{y^3}{3} - \epsilon t$$

How does the jump happen? We expect the imperfection  $s_1 - s_0$  to scale with  $\epsilon$  as  $\epsilon \rightarrow 0$ . Naturally, let the scaled time be  $s = \epsilon t$ .

$$\frac{d}{dt} = \epsilon \frac{d}{ds}$$

The ODE becomes:



$$\epsilon \frac{dY}{ds} = Y - \frac{Y^3}{3} - s$$

This is a **singular perturbation** problem!

Set  $\epsilon = 0 \implies s = Y_0 - \frac{Y_0^3}{3}$ .

*Sketch: The S-shaped curve again, highlighting the "outer layer" tracking the equilibrium curve, and "corner layers" where the rapid jump occurs.*

**Corner layer 1 thickness:** Let  $t_c = \frac{s-2/3}{\epsilon^\alpha}$ .

$$\frac{d}{ds} = \epsilon^{-\alpha} \frac{d}{dt_c}$$

Expand  $Y_c$ :

$$Y_c = 1 + \epsilon^b Y_{c1} + \epsilon^{2b} Y_{c2} + \dots$$

Substitute into the ODE. The left-hand side is  $O(\epsilon^{1-\alpha+b})$ . On the right-hand side, near the fold  $Y = 1$  and  $s = 2/3$ , we expand  $Y - Y^3/3 - s$ .

$$\epsilon^{1-\alpha+b} \frac{dY_{c1}}{dt_c} = -\epsilon^{2b} Y_{c1}^2 - \epsilon^\alpha t_c$$

Using a 3-term dominant balance:

$$1 - \alpha + b = 2b = \alpha$$

Solving gives  $b = 1/3$  and  $\alpha = 2/3$ . Thus, the delay  $\propto \epsilon^{2/3}$ .

**Corner layer ODE:**

$$\frac{dY_{c1}}{dt_c} = -Y_{c1}^2 - t_c$$

This is a Riccati equation. Trick: Let  $Y_{c1} = \frac{W'}{W}$ .

$$\frac{dW'}{dt_c} = \frac{WW'' - (W')^2}{W^2}$$

Substituting into the Riccati equation gives:

$$W'' = (-t_c)W \quad (\text{Airy equation})$$

$$W = c_0 \text{Ai}(-t_c) + c_1 \text{Bi}(-t_c)$$

$$Y_{c1} = \frac{-c_0 \text{Ai}'(-t_c) - c_1 \text{Bi}'(-t_c)}{c_0 \text{Ai}(-t_c) + c_1 \text{Bi}(-t_c)}$$

We determine  $c_0, c_1$  by matching with the outer solution as  $t_c \rightarrow -\infty$ . If  $c_1 \neq 0$ ,  $Y_{c1} \rightarrow -\frac{\text{Bi}'(-t_c)}{\text{Bi}(-t_c)}$  which has the wrong shape. Thus  $c_1 = 0$ , giving:

$$Y_{c1}(t_c) = -\frac{\text{Ai}'(-t_c)}{\text{Ai}(-t_c)}$$

This solution blows up (jumps) when  $\text{Ai}(-t_c) = 0$ . The first root is  $-t_c \approx -2.338 \implies t_c \approx 2.338$ .

$$2.338 = \frac{s_1 - 2/3}{\epsilon^{2/3}} \implies s_1 = s_0 + 2.338\epsilon^{2/3}$$

$$Y_c = 1 - \frac{\text{Ai}'(-t_c)}{\text{Ai}(-t_c)}\epsilon^{1/3}$$

## Forced Harmonic Oscillator

$$y'' + \omega_0^2 y = \cos(\omega_f t)$$

Homogeneous equation:

$$y_h'' + \omega_0^2 y_h = 0 \implies y_h = A \cos(\omega_0 t) + B \sin(\omega_0 t)$$

A particular solution for  $y_p'' + \omega_0^2 y_p = \cos(\omega_f t)$ : If  $\omega_0 \neq \omega_f$ :

$$y_p = \frac{\cos(\omega_f t)}{\omega_0^2 - \omega_f^2}$$

If  $\omega_0 = \omega_f$  (resonance), the amplitude grows linearly over  $t$ :

$$y_p = \frac{t}{2\omega_f} \sin(\omega_f t) \quad (\text{secular term})$$

A **secular term** typically appears when the RHS force is itself a solution of the homogeneous equation. Example:  $y'' - y = e^{-t}$  has a secular term because  $e^{-t}$  is a solution to  $y'' - y = 0$ .

## Nonlinear Spring and Duffing Oscillator

$$\ddot{y} + \delta \dot{y} + \alpha y + \beta y^3 = \gamma \cos(\omega t)$$

Terms:  $\delta \dot{y}$  = damping  $\alpha y$  = linear spring  $\beta y^3$  = nonlinear spring (Duffing term)  $\gamma \cos(\omega t)$  = external driving force

**Duffing oscillator with weak nonlinearity:**

$$y'' + y + \epsilon y^3 = 0$$

If  $\epsilon > 0$ , it is a hardening spring. If  $\epsilon < 0$ , it is a softening spring. B.C.:  $y(0) = 1$ ,  $y'(0) = 0$ .

With no damping and no forcing, we expect periodic behavior. Naively, this looks like a simple regular perturbation problem (we will show that this fails in an interesting way).

$$y = y_0 + \epsilon y_1 + \epsilon^2 y_2 + \dots$$

The ODE becomes:

$$(y_0'' + \epsilon y_1'' + \dots) + (y_0 + \epsilon y_1 + \dots) + \epsilon(y_0 + \epsilon y_1 + \dots)^3 = 0$$

$O(\epsilon^1)$  **Equation:**

$$y_1'' + y_1 + y_0^3 = 0 \implies y_1'' + y_1 = -y_0^3$$

Since  $y_0 = \cos t$ , we have:

$$y_1'' + y_1 = -\cos^3 t$$

Using the trigonometric identity:

$$\cos^3 t = \left( \frac{e^{it} + e^{-it}}{2} \right)^3 = \frac{1}{8}(e^{3it} + e^{-3it} + 3e^{it} + 3e^{-it}) = \frac{\cos 3t + 3\cos t}{4}$$

The equation can be separated into two parts:

$$y_a'' + y_a = -\frac{1}{4}\cos 3t$$

$$y_b'' + y_b = -\frac{3}{4}\cos t$$

Solving these gives:

$$y_a = a_1 \cos t + a_2 \sin t + \frac{1}{32} \cos 3t$$

$$y_b = b_1 \cos t + b_2 \sin t - \frac{3}{8}t \sin t \quad (\text{secular term})$$

Combining and applying the boundary conditions  $y_1(0) = 0$  and  $y_1'(0) = 0$ :

$$y_1(t) = A \cos t + B \sin t + \frac{1}{32} \cos 3t - \frac{3}{8}t \sin t$$

$$y_1(0) = 0 \implies A + \frac{1}{32} = 0 \implies A = -\frac{1}{32}$$

$$y_1'(0) = 0 \implies B = 0$$

$$\implies y_1(t) = -\frac{1}{32} \cos t + \frac{1}{32} \cos 3t - \frac{3}{8}t \sin t$$

**This is problematic because:**

- On physical grounds, the solution should be periodic, but the secular term breaks periodicity.
- Asymptoticity breaks down when  $t = O(1/\epsilon)$ . No matter how small  $\epsilon$  is, eventually the term  $\epsilon y_1$  becomes order  $O(1)$  and comparable to  $y_0$ .
- In fact, secular terms appear in all orders  $\implies$  we would need infinite terms to cancel the secular growth.

**Cause of the problem:** For  $y'' + y + \epsilon y^3 = 0$ , we expect the Duffing term to alter the frequency to be different from 1. But regular perturbation forces the zeroth-order solution to oscillate with  $\omega = 1$ , producing a secular term in the next order to compensate.

## Poincaré-Lindstedt Method

Expand not only  $y$ , but also  $\omega$ . Let  $\tau = \omega t$  (stretch time).

$$\omega = \omega_0 + \epsilon\omega_1 + \epsilon^2\omega_2 + \dots$$

When  $\epsilon = 0$ , the Duffing equation reduces to a simple harmonic oscillator, which gives  $\omega_0 = 1$ . The derivatives become:

$$\frac{d}{dt} = \frac{d}{d\tau} \frac{d\tau}{dt} = \omega \frac{d}{d\tau}$$

$$\frac{d^2}{dt^2} = \omega^2 \frac{d^2}{d\tau^2}$$

The ODE becomes:

$$\omega^2 Y'' + Y + \epsilon Y^3 = 0$$

With boundary conditions:  $Y(0) = 1$ , and  $\omega Y'(0) = 0 \implies Y'(0) = 0$  (with respect to  $\tau$ ).

Now expand  $Y = Y_0 + \epsilon Y_1 + \epsilon^2 Y_2 + \dots$

$$(\omega_0 + \epsilon\omega_1)^2 (Y_0'' + \epsilon Y_1'' + \dots) + (Y_0 + \epsilon Y_1) + \epsilon (Y_0 + \epsilon Y_1)^3 = 0$$

$O(\epsilon^0)$  **Equation:** Set  $\epsilon = 0$ .

$$\omega_0^2 Y_0'' + Y_0 = 0 \implies Y_0 = \cos\left(\frac{1}{\omega_0}\tau\right) = \cos\tau \quad (\text{since } \omega_0 = 1)$$

$O(\epsilon^1)$  **Equation:**

$$\omega_0^2 Y_1'' + Y_1 + 2\omega_0\omega_1 Y_0'' + Y_0^3 = 0$$

Since  $\omega_0 = 1$  and  $Y_0'' = -\cos \tau$ :

$$Y_1'' + Y_1 - 2\omega_1 \cos \tau + \cos^3 \tau = 0$$

$$Y_1'' + Y_1 = 2\omega_1 \cos \tau - \frac{\cos 3\tau + 3 \cos \tau}{4}$$

$$Y_1'' + Y_1 = -\frac{1}{4} \cos 3\tau + \left(2\omega_1 - \frac{3}{4}\right) \cos \tau$$

To **avoid resonance** (which generates secular terms), we must eliminate the  $\cos \tau$  term on the right-hand side:

$$2\omega_1 - \frac{3}{4} = 0 \implies \omega_1 = \frac{3}{8}$$

The equation then simplifies to:

$$Y_1'' + Y_1 = -\frac{1}{4} \cos 3\tau$$

The general solution is  $Y_1 = Y_{1h} + Y_{1p}$ :

$$Y_{1h}'' + Y_{1h} = 0 \implies Y_{1h} = A \cos \tau + B \sin \tau$$

$$Y_{1p} = \frac{-\frac{1}{4}}{-(3)^2 + 1} \cos 3\tau = \frac{1}{32} \cos 3\tau$$

$$Y_1 = A \cos \tau + B \sin \tau + \frac{1}{32} \cos 3\tau$$

Applying the ICs:  $Y_1(0) = 0$  and  $Y_1'(0) = 0 \implies B = 0$  and  $A + \frac{1}{32} = 0 \implies A = -\frac{1}{32}$ .

$$Y_1(\tau) = \frac{1}{32}(\cos 3\tau - \cos \tau)$$

Therefore, the uniform approximate solution is:

$$Y \approx Y_0 + \epsilon Y_1 = \cos \tau + \frac{\epsilon}{32}(\cos 3\tau - \cos \tau)$$

where  $\tau = \omega t = (1 + \frac{3}{8}\epsilon)t$ .

$$y(t) \approx \cos\left((1 + \frac{3}{8}\epsilon)t\right) + \frac{\epsilon}{32} \left[ \cos\left(3(1 + \frac{3}{8}\epsilon)t\right) - \cos\left((1 + \frac{3}{8}\epsilon)t\right) \right]$$

## Multiple Scale Analysis

In the Poincaré-Lindstedt method, the key idea is to introduce a stretched time variable  $\tau = \omega t$  where  $\omega = \omega_0 + \epsilon\omega_1 + \epsilon^2\omega_2 + \dots$ . Then  $Y_0(\tau) = \cos(\omega t) = \cos(\omega_0 t + \epsilon\omega_1 t + \epsilon^2\omega_2 t + \dots)$ . This suggests multiple time scales:

$$T_0 = t, \quad T_1 = \epsilon t, \quad T_2 = \epsilon^2 t$$

progressively taking a longer time  $(1, 1/\epsilon, 1/\epsilon^2)$  to take effect.

As an example, we use only 2 time scales  $T_0, T_1$ :

$$T_0 = t \quad (\text{fast time})$$

$$T_1 = \epsilon t \quad (\text{slow time})$$

**Key intuition:** The system behaves very differently on  $T_0$  and  $T_1$ . Treat  $T_0, T_1$  as independent variables.

$$Y(t) \rightarrow Y(T_0, T_1)$$

Using the chain rule:

$$\frac{d}{dt} Y(T_0, T_1) = \frac{\partial Y}{\partial T_0} \frac{dT_0}{dt} + \frac{\partial Y}{\partial T_1} \frac{dT_1}{dt} = \frac{\partial Y}{\partial T_0} + \epsilon \frac{\partial Y}{\partial T_1}$$

$$\frac{d^2 Y}{dt^2} = \left( \frac{\partial}{\partial T_0} + \epsilon \frac{\partial}{\partial T_1} \right)^2 Y = \frac{\partial^2 Y}{\partial T_0^2} + 2\epsilon \frac{\partial^2 Y}{\partial T_0 \partial T_1} + \epsilon^2 \frac{\partial^2 Y}{\partial T_1^2}$$

Expand  $Y = Y_0 + \epsilon Y_1 + \epsilon^2 Y_2 + \dots$  and substitute into the Duffing ODE:

$$\left( \frac{\partial^2}{\partial T_0^2} + 2\epsilon \frac{\partial^2}{\partial T_0 \partial T_1} + \dots \right) (Y_0 + \epsilon Y_1) + (Y_0 + \epsilon Y_1) + \epsilon(Y_0^3 + \dots) = 0$$

$O(\epsilon^0)$  **Equation:**

$$\frac{\partial^2 Y_0}{\partial T_0^2} + Y_0 = 0 \quad (\text{harmonic oscillator in } T_0)$$

$$\implies Y_0(T_0, T_1) = A(T_1)e^{iT_0} + A^*(T_1)e^{-iT_0}$$

This relates to the major motivation of 2-timing: to treat the cubic term, we write the solution as a fast oscillation  $e^{iT_0}$  modulated by a slow variation of amplitude  $A(T_1)$ .

$O(\epsilon^1)$  **Equation:**

$$\frac{\partial^2 Y_1}{\partial T_0^2} + Y_1 = -Y_0^3 - 2\frac{\partial^2 Y_0}{\partial T_0 \partial T_1}$$

Substitute  $Y_0 = Ae^{iT_0} + A^*e^{-iT_0}$ :

$$Y_0^3 = A^3e^{3iT_0} + 3A^2A^*e^{iT_0} + 3AA^{*2}e^{-iT_0} + A^{*3}e^{-3iT_0}$$

$$2\frac{\partial^2 Y_0}{\partial T_0 \partial T_1} = 2iA'e^{iT_0} - 2iA^{*'}e^{-iT_0}$$

To avoid secular growth, the resonant terms (coefficients of  $e^{iT_0}$  and  $e^{-iT_0}$ ) must vanish:

$$3A^2A^*e^{iT_0} + 2iA'e^{iT_0} = 0 \implies 3A^2A^* + 2iA' = 0$$

(The  $e^{-iT_0}$  term yields the complex conjugate:  $3AA^{*2} - 2iA^{*'} = 0$ ).

**Amplitude equation:** Write  $A(T_1) = R(T_1)e^{i\theta(T_1)}$ . Substitute into the amplitude equation:

$$2i(R'e^{i\theta} + iR\theta'e^{i\theta}) + 3R^3e^{i\theta} = 0$$

Divide by  $e^{i\theta}$ :

$$2iR' - 2R\theta' + 3R^3 = 0$$

Separate real and imaginary parts:

$$\begin{cases} -2R\theta' + 3R^3 = 0 \implies \theta' = \frac{3}{2}R^2, \\ 2R' = 0 \implies R = R_0 \text{ (constant)} \end{cases}$$

Thus,  $\theta(T_1) = \frac{3}{2}R_0^2T_1 + \theta_0$ .

$$A(T_1) = R_0e^{i(\frac{3}{2}R_0^2T_1 + \theta_0)}$$



Substituting back into  $Y_0$ :

$$Y_0(T_0, T_1) = Ae^{iT_0} + A^*e^{-iT_0} = 2R_0 \cos\left(T_0 + \frac{3}{2}R_0^2 T_1 + \theta_0\right)$$

Applying the Initial Conditions:  $y(0) = 1 \implies Y_0(0, 0) = 1 \implies 2R_0 \cos \theta_0 = 1$ .  $y'(0) = 0 \implies \frac{\partial Y_0}{\partial T_0}(0, 0) = 0 \implies -2R_0 \sin \theta_0 = 0 \implies \theta_0 = 0$ . So  $2R_0 = 1 \implies R_0 = 1/2$ . Finally:

$$Y_0 \approx \cos\left(t + \frac{3}{8}\epsilon t\right)$$

### Example 2: Weakly Damped Oscillator

$$y'' + 2\epsilon y' + y = 0$$

Here we have a spring with weak dissipation. Boundary Conditions:  $y(0) = 1$ ,  $y'(0) = 0$ . This has an exact solution (slightly shifted frequency, decaying amplitude):

$$y(t) = Ae^{-\epsilon t} \cos\left(\sqrt{1 - \epsilon^2}t + \phi_0\right)$$

If we try regular perturbation:

$$O(\epsilon^0) : y_0'' + y_0 = 0 \implies y_0 = \cos t$$

$$O(\epsilon^1) : y_1'' + y_1 = -2y_0' = 2 \sin t \quad (\text{resonance})$$

This yields a secular term:  $y_1 = -t \cos t$ .

$$y \approx \cos t - \epsilon t \cos t = (1 - \epsilon t) \cos t \approx e^{-\epsilon t} \cos t$$

Now use multiple scale analysis:  $T_0 = t$  (fast time scale),  $T_1 = \epsilon^\alpha t$  (slow time scale). Treat  $T_0, T_1$  as independent variables.

$$\frac{d}{dt} = \partial_0 + \epsilon^\alpha \partial_1$$

$$\frac{d^2}{dt^2} = \partial_0^2 + 2\epsilon^\alpha \partial_0 \partial_1 + \epsilon^{2\alpha} \partial_1^2$$

Substitute into the original ODE:

$$(\partial_0^2 + 2\epsilon^\alpha \partial_0 \partial_1 + \epsilon^{2\alpha} \partial_1^2)Y + 2\epsilon(\partial_0 + \epsilon^\alpha \partial_1)Y + Y = 0$$

Initial conditions:

$$Y(0, 0) = 1$$

$$(\partial_0 + \epsilon^\alpha \partial_1)Y(0, 0) = 0$$

Expand  $Y = Y_0 + \epsilon Y_1 + \dots$

$$(\partial_0^2 + 2\epsilon^\alpha \partial_0 \partial_1)(Y_0 + \epsilon Y_1) + 2\epsilon \partial_0(Y_0) + Y_0 + \epsilon Y_1 = 0$$

$O(\epsilon^0)$  **Equation:**

$$\partial_0^2 Y_0 + Y_0 = 0 \implies Y_0 = A(T_1) e^{iT_0} + c.c.$$

$O(\epsilon^1)$  **Equation:** To match the  $O(\epsilon)$  terms from the first derivative, we need  $\epsilon^\alpha = \epsilon \implies \alpha = 1$ .

$$\partial_0^2 Y_1 + Y_1 = -2\partial_0 \partial_1 Y_0 - 2\partial_0 Y_0$$

Evaluate the RHS:

$$\partial_0 Y_0 = iA e^{iT_0} - iA^* e^{-iT_0}$$

$$\partial_0 \partial_1 Y_0 = iA' e^{iT_0} - iA^{*'} e^{-iT_0}$$

We need to remove these resonant terms. For the  $e^{iT_0}$  component:

$$-2iA' - 2iA = 0 \implies A' + A = 0$$

**Amplitude Equation:**

$$A' + A = 0 \implies A(T_1) = A_0 e^{-T_1}$$

Apply Initial Conditions:  $Y_0(0, 0) = 1 \implies A_0 + A_0^* = 1$ .  $Y'(0) = 0 \implies \partial_0 Y_0(0, 0) = 0 \implies iA_0 - iA_0^* = 0 \implies A_0 = A_0^*$ . Thus,  $2A_0 = 1 \implies A_0 = 1/2$ .

$$Y_0 = \frac{1}{2}e^{-T_1}(e^{iT_0} + e^{-iT_0}) = e^{-T_1} \cos T_0 = e^{-\epsilon t} \cos t$$

### Example 3: Van der Pol Equation

$$y'' + \epsilon(y^2 - 1)y' + y = 0$$

- $|y| > 1$ : Dissipation  $\implies$  decay
- $|y| < 1$ : Excitation (energy injection)  $\implies$  grow

These balance to form a **limit cycle**. Note that the limit cycle amplitude  $\neq 1$ . When  $\epsilon = 0$ , it is simply a harmonic oscillator. When  $\epsilon \neq 0$ , the amplitude is selected by the balance of dissipation vs. excitation. Intuition: half of the cycle is dissipation, half of the cycle is excitation. **Q:** How does it approach the limit cycle? (Limit cycle size  $\sim 2$ ).

The fast oscillation and slow variation of amplitude give us two time scales:  $T_0 = t$ ,  $T_1 = \epsilon t$ .

$$\frac{d}{dt} = \partial_0 + \epsilon \partial_1, \quad \frac{d^2}{dt^2} = \partial_0^2 + 2\epsilon \partial_0 \partial_1 + \dots$$

Substitute into the ODE:

$$(\partial_0^2 + 2\epsilon \partial_0 \partial_1)Y + \epsilon(Y^2 - 1)(\partial_0 + \epsilon \partial_1)Y + Y = 0$$

$O(\epsilon^0)$  **Equation:**

$$\partial_0^2 Y_0 + Y_0 = 0 \implies Y_0 = A(T_1)e^{iT_0} + c.c.$$

$O(\epsilon^1)$  **Equation:**

$$\partial_0^2 Y_1 + Y_1 = -2\partial_0 \partial_1 Y_0 - (Y_0^2 - 1)\partial_0 Y_0$$

Isolate the resonant terms (proportional to  $e^{iT_0}$ ):

$$\partial_0 Y_0 = iAe^{iT_0} - iA^*e^{-iT_0}$$

$$Y_0^2 = A^2 e^{2iT_0} + 2AA^* + A^{*2} e^{-2iT_0}$$

Multiplying these out, the coefficient of  $e^{iT_0}$  from  $(Y_0^2 - 1)\partial_0 Y_0$  is  $iA(AA^* - 1)$ . Setting the resonant terms to zero:

$$-2iA' - iA(AA^* - 1) = 0 \implies 2A' = -A(AA^* - 1)$$

Write  $A(T_1) = R(T_1)e^{i\theta(T_1)}$ :

$$A' = R'e^{i\theta} + iR\theta'e^{i\theta}$$

Substitute into the amplitude equation:

$$2(R' + iR\theta')e^{i\theta} = -Re^{i\theta}(R^2 - 1)$$

$$2R' + 2iR\theta' = -R(R^2 - 1)$$

Separate real and imaginary parts:

*Sketch: Phase plot for the real part  $R'$ . It shows a 1D nonlinear dynamical system where  $R = 0$  is an unstable fixed point and  $R = 1$  is a stable fixed point. The flow arrows point from 0 to 1, and from infinity back to 1.*

*Sketch: Bifurcation diagram showing the stable branch  $R = 1$  emerging symmetrically from the origin, resembling a pitchfork, indicating a stable limit cycle of amplitude 2.*

**Imaginary Part:**

$$2R\theta' = 0 \implies \theta' = 0 \implies \theta = \theta_0$$

**Real Part:**

$$2R' = -R(R^2 - 1) \implies \frac{2R'}{R^3} = \frac{1}{R^2} - 1$$

Let  $u = \frac{1}{R^2}$ , then  $u' = -\frac{2R'}{R^3}$ . The equation becomes:

$$-u' = u - 1 \implies (u - 1)' = -(u - 1)$$

$$\implies \frac{1}{R^2} - 1 = Ce^{-T_1}$$

At  $T_1 = 0$ ,  $C = \frac{1}{R_0^2} - 1$ .

$$R(T_1) = \frac{1}{\sqrt{1 + \left(\frac{1}{R_0^2} - 1\right) e^{-T_1}}}$$

The leading order solution is:

$$Y_0 = 2R(T_1) \cos(T_0 + \theta_0) = \frac{2 \cos(t + \theta_0)}{\sqrt{1 + \left(\frac{1}{R_0^2} - 1\right) e^{-\epsilon t}}}$$

Notice that as  $t \rightarrow \infty$ ,  $e^{-\epsilon t} \rightarrow 0$ , so the denominator goes to 1. Thus,  $Y_0 \rightarrow 2 \cos(t + \theta_0)$ , meaning the limit cycle amplitude approaches 2, exactly as predicted by the 1D dynamics of  $R$ .